

Experimental validation of a causal local corrector on top of a variational nominal trajectory for collision avoidance among moving ground robots

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Abstract. Safe navigation among independently moving agents requires global progress and reaction without access to their future motion. This study evaluates an architecture that separates a variational nominal trajectory, computed offline from the static scene, from a causal local corrector driven online by observed agent states. Two independent studies comprised a simulation comparison of twelve combinations of nominal generators and local correctors, each calibrated separately, and an experimental evaluation of one combination on an encoderless differential-drive rover using overhead vision as its sole feedback. At a 10 cm terminal tolerance, the two leading variational combinations achieved joint safety-and-arrival rates of 65.38 % and 63.46 %, compared with 48.08 % for the best straight-line combination. In forty physical trials spanning two repeated configurations with one or two moving agents, the rover reached the goal every time, with no contact or negative clearance; the smallest agent margin was 6.3 cm, and the loop operated near 20 Hz. Paired simulation reproduced path geometry to about 5 cm mean departure but not path length. These results support feasibility in the tested configurations, without establishing a formal separation guarantee or general performance superiority.

Keywords: Trajectory optimization / Collision avoidance / Differential-drive robots / Causal control / Experimental validation / Vision-based control

Nomenclature

Trajectory and cost

$\mathbf{x}$	robot position, m	V	total potential
$\mathbf{x}_0$	initial position, m	V_g	goal potential
$\mathbf{x}_g$	goal position, m	V_o	obstacle potential
$\mathbf{x}_{ref}$	nominal trajectory, m	V_w	wall potential
T	planning horizon, s	V_a	agent potential
N	discretisation intervals	β	goal weight
Δt	time step, s	γ	obstacle weight
J	continuous action	λ_s	smoothing weight
J_h	discrete action	α	kinetic weight
$g_{o,k}$	obstacle repulsion gain	R_w	wall influence, m
$\mathbf{c}_k$	obstacle centre, m	r_k	obstacle radius, m

Geometry, correction and evaluation

d	edge clearance, m	R	influence radius, m
r_{rob}	robot safety radius, m	$\mathbf{F}$	total force
$\mathbf{v}$	corrector velocity, m/s	m	robot mass, kg
k_a	attraction gain	k_m	moving-agent gain
k_s	static gain	k_b	boundary gain
c_d	damping coefficient	r_p	perception radius, m
t_ℓ	lookahead, s	v_{max}	speed bound, m/s
a_{max}	acceleration bound, m/s ²	α_{cmd}	command filter
H	homography	τ	terminal tolerance, m
k_π	parallax factor	I_s, I_g, I_j	evaluation indicators
h_m	marker height, m	h_c	camera height, m

1 Introduction

Ground robots increasingly operate in shared environments. Intralogistics vehicles, inspection platforms and collaborative cells share their floor with people, handling equipment and other robots, none of which negotiates its path with them or announces where it intends to go. The motion problem this creates has two time scales that do not reconcile easily. Over the length of a mission the robot must choose a route and progress towards a goal under criteria of path length, smoothness or energy; over a fraction of a second it must react to an agent that was not part of the scene when that route was computed. The two are further separated by the vehicle itself: a differential-drive rover cannot instantaneously adopt an arbitrary velocity direction, so an evasive command that is admissible for a point mass is not necessarily executable by the machine that has to perform it [1]. Simulation does not settle the question either, since slip, actuator dynamics, latency and perception noise are precisely what a physical testbed exposes and a model omits [2].

Continuous trajectory optimisation is well suited to the long-horizon planning problem. Direct methods transcribe a continuous motion problem into a finite-dimensional numerical one [3], and the robotics literature has produced several mature instances of that idea: covariant gradient refinement of an initial path [4], stochastic perturbation when no explicit cost gradient is available [5], and sequential convex approximation with

continuous collision checking [6]. The variational formulation used in this work descends from a model-based and data-driven hybrid treatment of the same optimal planning problem [7], in which part of the computational cost is deliberately moved offline. What all of these approaches share, however, is that the trajectory they return is a function of the scene supplied to the optimiser. An agent that was not in that scene is not handled by the plan, and recomputing the plan does not by itself solve the problem, because the information the recomputation would need does not exist yet.

That last point motivates the present work. It is important to distinguish a planner that uses observed agent states from a global optimiser that evaluates the position of a moving agent at the future instants of its own horizon. Doing so in simulation produces excellent trajectories and is entirely meaningless, since the quantity being read is the scenario's own script rather than an observation. We use the term *causal* throughout this paper in the restricted sense defined in Section 2.3: a component is causal when it accesses only the current and past states of the moving agents. This is not a claim of sophistication — a prediction built from available observations is causal too, and so is a receding-horizon controller — but a minimal admissibility condition. No exact future agent trajectory is supplied to any component of the architecture reported here, and the property is enforced by construction rather than by convention.

The short-range half of the problem has been addressed by three families of methods whose trade-offs are well documented. Superimposed attractive and repulsive fields give a fast, continuous and inexpensive correction [8], at the cost of structural weaknesses that were characterised early: local minima, oscillation in narrow passages and loss of progression [9]. Navigation functions show that spurious minima can be eliminated, but within a structured class of spaces and obstacles whose assumptions do not transfer automatically to an empirical field acting on moving agents [10]. Within the same reactive family, sampling admissible commands rather than summing forces — through a directional histogram [11] or a dynamically feasible velocity window [12] — accounts for acceleration limits explicitly, while remaining local and sensitive to its scoring weights. A second family adds geometric anticipation: velocity obstacles identify the commands that would lead to a collision under an assumed motion model [13], and their reciprocal formulations distribute the avoidance effort between agents [14, 15]. The absence of explicit communication in those formulations should not be read as an absence of cooperation: reciprocity is a behavioural assumption about the other agent, and it is exactly the assumption that a non-cooperative agent breaks. A third family, receding-horizon optimal control, integrates the non-holonomic kinematics, the actuator saturations and the clearance constraints directly [16, 17]; its horizon is a model-based extrapolation of available observations, not knowledge of the true future, and its performance depends on prediction error and on the online computational budget.

Combining a global reference with a local corrector is therefore not, in itself, a new architecture, and it would be indefensible to present it as one. Reciprocal avoidance has been adapted to non-holonomic robots and validated on physical multi-robot teams [18]; a preferred motion issued by a global planner has been corrected over a short horizon under non-holonomic constraints on several platforms [19]; two-layer combinations of a potential field with a local velocity-space planner facing moving obstacles have been published [20]; avoidance has been extended to agents whose degree of cooperation is unknown and estimated from observation [21], and then to differential-drive robots facing cooperative and non-cooperative agents with causal state estimation [22]. Offline precomputation combined with real-time execution and external visual localisation has likewise been demonstrated, although on an omnidirectional robot and in an essentially static environment [23]. What remains sparsely documented is not the decomposition but a specific instantiation of it, evaluated rather than asserted: a variational nominal trajectory computed offline on the static scene alone, corrected online from the currently observed positions of non-cooperative moving agents, and executed on an encoderless differential-drive rover whose only feedback is an overhead camera. Two methodological conditions govern whether such an evaluation means anything. The comparison of candidate architectures must not be biased by tuning, since the ranking of planners is known to depend on their parameters and on the pipeline that surrounds them [24, 25]; and the success criterion must require progression and safety simultaneously, because a robot that never moves does not collide and a robot that arrives may have violated its margin [26]. Neither of these is a substitute for a proof: a guarantee of collision avoidance requires a model, explicit assumptions and a formal argument [27], and no such

guarantee is claimed here.

The question this paper addresses is consequently an experimental one. On the scenarios and on the testbed studied, what does it cost to separate an offline variational plan from a strictly causal local correction — in safety margin, in terminal accuracy and in fidelity to the plan — and does the resulting chain hold a usable real-time budget on hardware? Three hypotheses are tested. **H1**, in simulation: once every candidate architecture is tuned in its own parameter space and under a common budget, the choice of the nominal–corrector pair still changes joint success substantially, and the resulting ordering is not the one that a safety rate alone would produce. **H2**, in simulation: the global plan improves joint success over its own ablation, by an amount that depends on the terminal requirement rather than being a fixed property of the architecture. **H3**, on hardware: the complete perception–decision–command chain reaches the goal without physical contact in a repeated one-agent and two-agent protocol, while sustaining the requested loop rate. **H3** is a feasibility hypothesis over the domain tested; it is not a safety guarantee, and it says nothing about the architectures that were not run on hardware.

The contributions are correspondingly delimited. First, the causal separation itself is specified and enforced by construction, the nominal scene being built without moving agents and the corrector never evaluating an agent state later than the present one. Second, twelve nominal-generator/corrector combinations are compared in simulation under a protocol in which each combination calibrates its own coefficients, none of them inherited from another, and in which success requires safety and arrival jointly; that comparison also shows that a safety rate reported alone selects the least useful architecture of the study. Third, the retained combination is validated end to end on a physical testbed over forty runs of two frozen tasks, one with a single moving agent and one with two: the goal was reached in every run, neither a collision nor a negative clearance was recorded, and the closed loop held its requested rate at a median of 19.84 Hz or above. Fourth, the paired simulation is assessed rather than assumed: it reproduces the executed path to a mean departure of about 5 cm and does not reproduce its length, which is reported as a partial result and analysed as such. All quantitative claims are restricted to the reported campaign, to its testbed and to at most two moving agents.

The paper is organised as follows. Section 2 states the motion task, the joint success criterion and the causality condition, and gives an overview of the processing chain. Section 3 describes the experimental platform, the metric vision chain, the offline nominal generator, the causal corrector, the wheel-level command, the calibration protocol and the evaluation metrics. Section 4 reports the simulation comparison, the real-time budget, the physical campaign and the simulation-versus-experiment agreement. Section 5 interprets these results and states the limitations, and Section 6 concludes.

2 Problem statement and system overview

2.1 Motion task

A single differential-drive ground robot must travel from an initial position $\mathbf{x}_0 \in \mathbb{R}^2$ to a goal position $\mathbf{x}_g \in \mathbb{R}^2$ inside a bounded planar arena. The arena contains two classes of obstruction. Fixed obstacles are rigid columns whose positions are known before the run and remain unchanged throughout it.

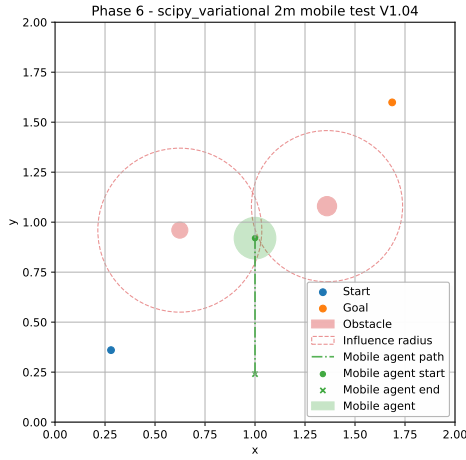


Figure 1. Task geometry. The controlled robot travels from $\mathbf{x}_0$ to $\mathbf{x}_g$ inside the instrumented arena, among fixed obstacle columns and independently moving ground robots. Dashed circles indicate the conservative bounding discs used to define the edge-to-edge clearance of Equation (1).

Moving agents are other ground robots which travel through the same arena during the run, which are not controlled by the robot under study, and whose future positions are never made available to it.

The controlled robot and every obstruction are modelled as discs in the horizontal plane. All distances used by the planner, by the corrector and by the evaluation metrics are *edge-to-edge clearances*: for an obstruction k of centre $\mathbf{c}_k$ and radius r_k ,

$$d_k(\mathbf{x}) = \|\mathbf{x} - \mathbf{c}_k\| - r_k - r_{\text{rob}}, \quad (1)$$

where r_{rob} is the safety radius of the controlled robot. A clearance of zero therefore means that the two conservative bounding discs touch, and a negative clearance means that they overlap. Because both radii are conservative envelopes rather than exact chassis outlines, a negative clearance does not by itself indicate physical contact. This distinction is maintained throughout the paper.

Figure 1 shows the geometry of the task, the two classes of obstruction, the safety radii and the metric reference frame.

2.2 Success criterion

The task carries two simultaneous requirements. The robot must remain *safe*, that is avoid every fixed obstacle, every moving agent and the arena boundary, and it must *terminate near the goal*. Neither requirement is sufficient on its own, and this is not a formal remark: each one, taken alone, admits a degenerate optimum. A policy that minimises only the final distance to the goal is free to drive straight through an obstacle, whereas a policy that maximises only the clearance is best served by a robot that never moves. Reporting either quantity in isolation therefore rewards a behaviour that the task does not want.

Both requirements are consequently combined into a single

binary outcome per run. For a run r and a terminal tolerance τ ,

$$I_s(r) = \begin{cases} 1 & \text{if no collision, no arena exit} \\ & \text{and no execution error occurred,} \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

$$I_g(r; \tau) = \begin{cases} 1 & \text{if } \|\mathbf{x}_r(T_r) - \mathbf{x}_g\| \leq \tau, \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

$$I_j(r; \tau) = I_s(r) I_g(r; \tau), \quad (4)$$

where T_r is the duration of run r . The joint indicator I_j is the quantity this work seeks to maximise, and rates computed from it satisfy $R_j \leq \min(R_s, R_g)$ by construction. The evaluation protocol built on these indicators is detailed in Section 3.7.

2.3 Causality

The term *causal* is used in this paper in a single, restricted and verifiable sense:

A component is causal when it accesses only the current and past states of the moving agents. Their future state is never queried.

This definition is deliberately narrow. It is the property that separates a controller which can be transferred to hardware from one which cannot, and it is enforced by construction rather than by convention: the scene used to compute the nominal trajectory excludes moving agents entirely, and the corrector never evaluates an agent position at a time later than the current one. The mechanisms that guarantee this are described in Section 3.4.3.

Causality in this sense does not mean that the model is complete. A causal controller may still be optimistic about sensing: assuming that the current state of every agent is available without delay, loss or uncertainty is itself an idealisation. The experimental campaign reported in this paper is subject to those effects, and the differences they produce are examined rather than assumed away.

2.4 System overview

The complete chain is organised in seven stages, split between an offline part and a real-time part. The boundary between the two is the point at which causality becomes a constraint.

1. **Define.** The task specification — start, goal, arena bounds, fixed obstacles, kinematic limits — is assembled into a static world description. Moving agents are deliberately absent from it.
2. **Initialise.** A family of parametric candidate paths is generated and evaluated, and the best candidate is retained as a warm start.
3. **Optimise.** A nominal trajectory $\mathbf{x}_{\text{ref}}(t)$ is obtained by minimising a discrete action functional over the static scene. This stage is performed offline.
4. **Correct.** At each control step, a local corrector combines an attraction towards a look-ahead point on $\mathbf{x}_{\text{ref}}$ with repulsions from the currently observed moving agents, the fixed obstacles and the arena boundary, together with viscous damping. This stage is real-time and causal.

5. **Command.** The corrected velocity and heading are converted into left and right wheel commands, saturated and smoothed, then transmitted to the robot.
6. **Observe.** An overhead industrial camera provides the metric pose of every marker in the arena, closing the loop.
7. **Measure.** A complete log of every run supports the post-processing metrics: clearances, tracking error, final distance, stopping reason and timing budget.

Stages 4 to 6 form the closed real-time loop. Figure 2 shows the full chain and the offline / real-time boundary.

The architecture rests on a single design decision: what can be optimised globally is optimised offline on the part of the scene that is known and static, and what cannot be known in advance is handled online by a local correction that never looks ahead of the present. The remainder of this paper describes that architecture in detail and reports what it costs, in measurable terms, on a physical testbed.

3 Materials and methods

This paper reports two complementary studies. The first, described in Section 3.6 and reported in Section 4.1, is a systematic comparison of twelve solver-controller architectures under an unbiased automatic calibration protocol, carried out entirely in simulation. The second, described in Sections 3.1 to 3.5 and reported in Sections 4.2 to 4.4, is the experimental validation of one such architecture on a vision-in-the-loop testbed. The coefficient set used in the physical trials was tuned on the hardware and is not the one produced by the calibration study; the two are reported independently.

3.1 Experimental platform

3.1.1 Arena

Experiments are carried out on a printed polyvinylchloride mat of $2.50\text{ m} \times 2.50\text{ m}$ carrying a regular grid. The mat is flexible, so it is laid on a rigid floor of 15 mm plywood panels cut to tile the same $2.50\text{ m} \times 2.50\text{ m}$ footprint. This substructure is not a mounting detail: the robots run on small unsprung wheels and, having no wheel encoders, possess no proprioceptive means of detecting or compensating a locally uneven surface. The plywood floor is what keeps the driving plane coincident with the metric plane recovered by the homography.

Four fiducial markers define the metric reference frame. These markers were initially placed close to the edges of the mat, where residual lens distortion is strongest, and were subsequently moved 25 cm inwards from each edge. Their centres now define the metric frame, so that the usable arena is

$$2.50\text{ m} - 2 \times 0.25\text{ m} = 2.00\text{ m}$$

in each direction. This trades experimental surface for a localisation accuracy that remains uniform over the whole working area; the measured benefit is quantified in Section 3.2. The origin of the metric frame is the centre of the bottom-left calibration marker, with x pointing right and y pointing up in the camera view, and the calibration offset is therefore zero.

3.1.2 Robots

Four identical commercially available four-wheel differential-drive platforms are used: one *controlled robot*, whose motion is the object of this study, and up to three *moving agents* acting as

independent disturbances. Each platform is driven by brushed gear motors *without wheel encoders*, and is supervised by an on-board microcontroller running unmodified factory firmware. The absence of encoders is a structural feature of the setup rather than an incidental limitation: *the only feedback available to the control loop is the external overhead camera*, and no local wheel-level closed loop exists anywhere in the chain.

The robots are represented by a conservative bounding disc of radius $r_{\text{rob}} = 0.130\text{ m}$, slightly larger than the measured half-diagonal of the chassis.

Fixed obstacles are additively manufactured columns of square cross-section, each carrying a fiducial marker on its top face; they are visible in Figure 3. That marker is functional rather than decorative. It allows the pose of every obstacle to be measured by the same vision chain that tracks the robots, instead of being entered by hand into a scenario file, and it makes each obstacle individually identifiable to an observer watching the arena or the recorded video. The obstacle marker planes stand 0.285 m above the mat for two of the columns and 0.340 m for the third. Each obstacle is modelled with a radius of 0.10 m. Their influence region is specified as a centre-to-centre distance of 0.30 m, which corresponds to an influence radius of 0.07 m once expressed on the edge-to-edge clearance of Equation (1); the latter is the value that enters the potential of Equation (11).

The kinematic envelope applied to the controlled robot during the reported BIS experiments is 0.25 m/s in speed and 1.0 m/s^2 in acceleration. These bounds are deliberately conservative and were chosen for three reasons. At low speed the avoidance manoeuvres remain legible both in the logs and in the recorded video; a margin against collision is preserved in a 2 m arena shared by three robots; and the low-level speed response of the controller board became less repeatable at higher command levels, an effect that is quantified in Section 3.5.

3.1.3 Perception and computing hardware

A colour industrial camera operated in Mono8 is mounted overhead on an adjustable boom stand, with its optical centre 2.07 m above the mat, and acquires images at a resolution of 1500×1200 pixels. The camera is cabled to a wireless access point at the edge of the arena, which relays the stream over a 5 GHz link to a single-board computer *carried on the controlled robot*, which runs the complete real-time loop — image processing, causal correction and wheel command generation — and drives the motor controller through a short onboard serial link. The robot therefore carries its own computing and power supply; the overhead camera is the only sensor, and it is the only element of the measurement chain external to the robot. The moving agents are driven over an independent network link, described in Section 3.1.4.

That wireless hop is the sizing constraint of the whole link, and it is worth stating quantitatively because it is easily underestimated. The stream is not compressed: one 8-bit Mono8 frame of 1500×1200 pixels is 1.8 MB, so 20 Hz acquisition sustains a payload of 288 Mbit/s for the whole duration of a run. The camera is therefore wired, over its 5GigE interface, to a tri-band Wi-Fi 7 access point placed at the edge of the arena, and the robot does not rely on the radio integrated in the single-board computer but on an external USB 3.0 Wi-Fi 7 adapter. The logs of the reported campaign record a negotiated link rate of 1201 Mbit/s in the 5 GHz band, so the stream occupies about a quarter of the available capacity. That margin, rather than the

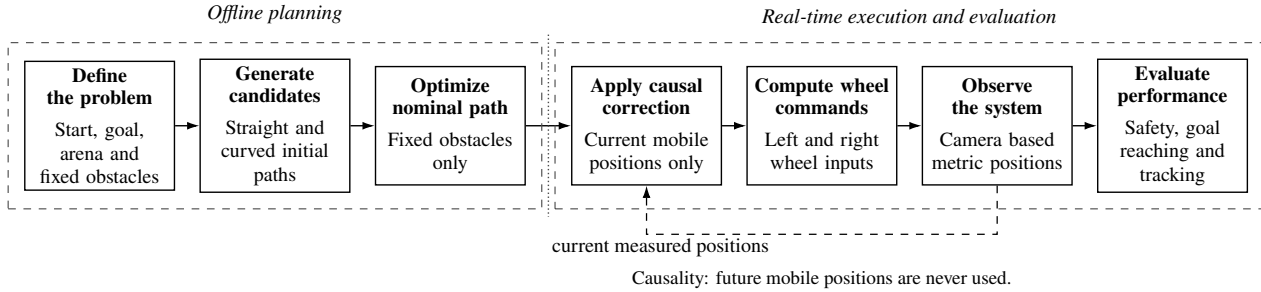


Figure 2. End-to-end processing chain. Stages 1 to 3 are executed offline on the static scene; stages 4 to 6 form the real-time closed loop and only access the current state of the moving agents. Stage 7 is post-processing.



Figure 3. Experimental testbed: instrumented arena, controlled robot and moving agents carrying their fiducial markers, the printed fixed-obstacle columns with their own markers on the top face, and the overhead camera mount.

nominal rating of any single component, is what makes a 20 Hz closed loop over a wireless link practicable.

The single-board computer is fitted with an active cooler, that is, a heatsink and a fan, and boots from a 256 GB memory card that also holds the raw log of every run. The cooler is a deliberate precaution rather than an afterthought: the vision stage occupies four threads continuously at 20 Hz, and any thermal throttling of the board under that sustained load would couple the timing figures reported in Section 4.2 to how long the board had been running.

Table 1 summarises the platform specification, and Figure 3 shows the assembled testbed.

Every active element of the testbed — camera, computer, robots, network — is a commercially available component, and no electronic part was designed or modified for this work. The only items made specifically for the bench are two passive mechanical ones: the plywood floor supporting the mat and the printed obstacle columns.

3.1.4 Moving agents

The moving agents are driven independently over a local network link, each receiving its own trajectory and speed setpoints before the run begins. No command is issued during the run. *The agents have no knowledge of the controlled robot and perform*

no avoidance manoeuvre of their own: all avoidance behaviour reported in this paper is produced by the controlled robot alone. This is the situation the paper addresses, and it differs from a coordinated multi-robot demonstration.

Each agent is placed manually at its starting position and commanded to move in a straight line. Because the agents also run open-loop without wheel encoders, their realised paths deviate laterally from the intended straight line: over the reported campaign the ratio of net displacement to travelled path length — unity for a perfect straight line — averages 0.835, so the realised paths exceed the straight line by about 20 %. This drift is consistent with the left–right asymmetry identified on the same platform in Section 3.5. Within a block, the agent placements and speed setpoints are held fixed, so that the runs repeat one disturbance configuration rather than sampling several.

3.2 Vision and metric localisation

3.2.1 Processing chain

Localisation is entirely external. Every pose used by the control loop is obtained from the overhead camera through the following chain:

1. acquisition of a 1500×1200 Mono8 image;
2. downscaling from 1500×1200 to 1000×800 ;
3. square fiducial marker detection [29] with sub-pixel corner refinement;
4. lens distortion correction using the intrinsic calibration;
5. planar homography mapping corrected pixel coordinates to metres;
6. parallax correction for markers that do not lie in the mat plane;
7. forward-axis offset correction, mapping marker heading to robot heading.

The output is a metric pose (x, y, θ) per marker identifier. Markers are assigned by role: four to the corners of the metric frame, one to the controlled robot, one to each moving agent, and one to each fixed obstacle. Marker planes are at different heights above the mat — the controlled robot marker at 158.5 mm, the agent markers at 90 mm — which is what makes the parallax correction necessary.

Table 1. Experimental platform specification. The reported robot mass is that of the complete assembly used in the runs, with all electronics, power supply and fixtures fitted.

Item	Value
Printed mat	2.50 m × 2.50 m
Mat substructure	15 mm plywood floor, same footprint
Usable arena (metric frame)	2.00 m × 2.00 m
Robots	4 differential-drive platforms, no wheel encoders
Robot safety radius r_{rob}	0.130 m
Controlled robot mass, as run (measured)	1.2588 kg
Speed / acceleration bound	0.25 m/s / 1.0 m/s ²
Fixed obstacles	Printed square-section columns, marked
Fixed obstacle radius (model)	0.10 m
Fixed obstacle influence (centre-to-centre)	0.30 m
Fixed obstacle influence (edge-to-edge)	0.07 m
Moving-agent influence (edge-to-edge)	0.60 m
Camera	Basler ace 2 a2A1920-165g5cBAS, 5GigE
Acquisition format	1500 × 1200, 8-bit Mono8
Uncompressed stream rate	1.8 MB per frame, 288 Mbit/s
Lens	3.5 mm fixed focal length
Camera cooling	Passive heat sink
Camera optical centre height	2.07 m
Access point	TP-Link Archer BE550, Wi-Fi 7
Robot network adapter	NETGEAR Nighthawk A9000, USB 3.0
Real-time computer (carried on the robot)	Raspberry Pi 5
Computer cooling	Raspberry Pi Active Cooler
Computer storage	256 GB microSD card
Vision library	OpenCV 5.0.0 [28], 4 threads
Marker dictionary	ArUco DICT_4X4_50

3.2.2 Intrinsic calibration

Intrinsic parameters are estimated by planar calibration [30] from twenty valid views of a printed chessboard-and-marker calibration board, giving a global root-mean-square reprojection error of 0.191 px. The resulting parameters are stored with a cryptographic digest and were kept unchanged for the whole campaign.

3.2.3 Homography

The mapping from image plane to mat plane is a planar homography estimated by the direct linear transformation and solved by singular value decomposition [31], using the four corner markers as control points. Writing $\tilde{\mathbf{x}}_{px}$ for the homogeneous distortion-corrected pixel coordinates and $\tilde{\mathbf{x}}_m$ for the homogeneous metric coordinates,

$$\tilde{\mathbf{x}}_m = \mathbf{H} \tilde{\mathbf{x}}_{px}, \quad \mathbf{x}_m = \frac{1}{\tilde{x}_{m,3}} \begin{pmatrix} \tilde{x}_{m,1} \\ \tilde{x}_{m,2} \end{pmatrix}. \quad (5)$$

Because the homography is estimated on already undistorted coordinates, the two error sources — lens distortion and projective mapping — are handled separately rather than being absorbed into a single fit.

3.2.4 Parallax correction

A marker lying at a height h_m above the mat is not imaged at the position of its ground projection. After lens distortion correction and assuming a pinhole camera model with the optical axis approximately normal to the mat, the parallax reduces to a radial scaling about the principal point. By similar triangles,

the corresponding scale factor is

$$k_\pi = 1 - \frac{h_m}{h_c}. \quad (6)$$

For the controlled robot, $h_m = 0.1585$ m and $h_c = 2.07$ m, giving $k_\pi = 0.9234$. The correction is applied only to raised markers; the four calibration markers lie in the mat plane and therefore require no correction. The marker heights were measured mechanically: the lower and upper edges of the controlled robot marker plane were found at 157 mm and 160 mm respectively, giving a centre height of 158.5 mm.

3.2.5 Forward-axis offset

The marker frame does not coincide with the driving axis of the robot, and the constant angle between them must be identified. Four runs with imposed headings (0° , $+90^\circ$, 180° , -90°) were recorded, and the offset was taken as the *circular mean* of the difference between expected heading and raw marker yaw. Circular statistics are used throughout for this angular quantity. The identified offset is 1.596 73 rad (91.486°), with a circular dispersion between runs of 1.316°, which also bounds the residual heading error after correction. Marker detection was complete over all four identification runs.

3.2.6 Localisation accuracy

Table 2 reports the accuracy of the chain at each stage of its construction, measured at nine known ground positions and at five known positions on the robot.

Distortion correction reduces the mean ground error by 84 %, with individual improvements ranging from 66 % to 89 % and no position degraded. Parallax correction reduces the mean error

Table 2. Localisation accuracy through the successive stages of the vision chain. Ground positions are measured with a marker lying in the mat plane; robot positions are measured with the raised marker of the controlled robot.

Stage	Mean error	Maximum error	Detection
Homography only, no distortion correction (ground)	36.3 mm	45.2 mm	—
Intrinsic calibration and recomputed homography (ground)	5.8 mm	9.0 mm	100 %
Raised marker, before parallax correction (robot)	60.8 mm	78.8 mm	—
After parallax correction (robot)	11.6 mm	23.1 mm	100 %

on the raised marker of the controlled robot by approximately 81 %. The final mean localisation error of 11.6 mm is 5.43 times smaller than the smallest moving-agent margin reported later in the paper (63 mm). The worst localisation error of 23.1 mm is 12.86 times smaller than the smallest reported static-obstacle margin (297 mm). These comparisons are discussed in Section 5.5.

Figure 4 maps the residual error before and after correction.

Figure 5 shows a raw camera frame together with the annotated detection output.

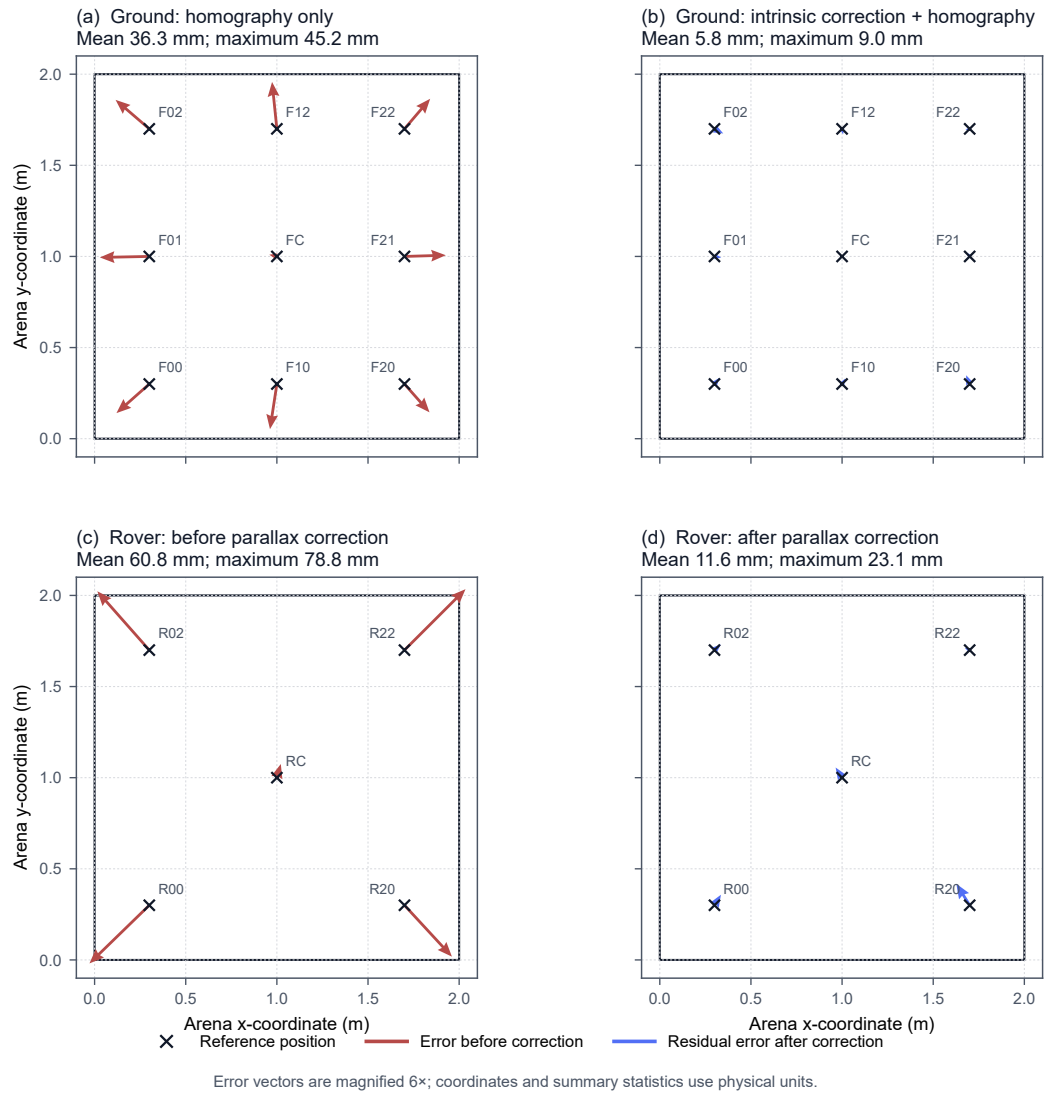


Figure 4. Localisation error before and after calibration, at nine known ground positions and five known positions on the controlled robot.

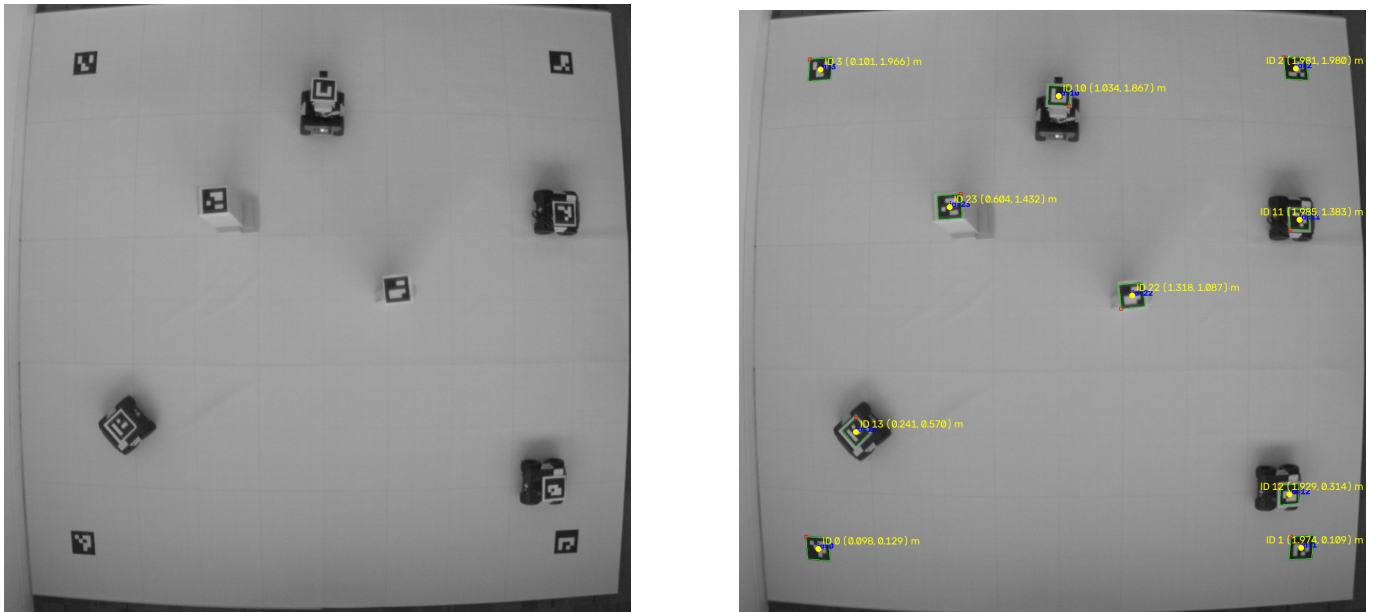


Figure 5. Overhead camera view. Left: raw acquired frame. Right: detected markers with identifiers, metric frame and reconstructed poses.

3.3 Offline nominal trajectory generation

Four nominal generators and three causal correctors are described in this section and the next. They constitute the common method catalogue of the two studies reported in this paper. *The experimental campaign of Section 4.3 uses the variational generator combined with the semi-implicit Euler corrector*; the remaining combinations are compared in Section 4.1.

3.3.1 Continuous formulation

The nominal trajectory is obtained by minimising an action functional over planar paths with fixed endpoints [32]. For a path $\mathbf{x} : [0, T] \rightarrow \mathbb{R}^2$,

$$J[\mathbf{x}] = \int_0^T \mathcal{L}(\mathbf{x}, \dot{\mathbf{x}}, t) dt, \quad \mathcal{L} = \frac{1}{2} \alpha \|\dot{\mathbf{x}}(t)\|^2 + V(\mathbf{x}(t)), \quad (7)$$

subject to the hard boundary conditions $\mathbf{x}(0) = \mathbf{x}_0$ and $\mathbf{x}(T) = \mathbf{x}_g$. The stationarity condition gives

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{x}}} \right) - \frac{\partial \mathcal{L}}{\partial \mathbf{x}} = 0 \quad \Rightarrow \quad \alpha \ddot{\mathbf{x}} - \nabla V(\mathbf{x}) = 0. \quad (8)$$

It is important to state explicitly that $\mathcal{L}$ in Equation (7) is a *cost* Lagrangian and not the mechanical Lagrangian $T - V$: the potential enters with a positive sign, and α is a weight on path smoothness rather than a mass. A second-order formulation — no $\ddot{\mathbf{x}}$ in $\mathcal{L}$ — is retained for numerical stability, for compatibility with the real-time budget, and because Equation (8) then admits a direct mechanical reading.

3.3.2 Potential

The potential is additive over the elements of the *static* scene:

$$V(\mathbf{x}) = V_g(\mathbf{x}) + V_o(\mathbf{x}) + V_w(\mathbf{x}) + V_a(\mathbf{x}), \quad (9)$$

with a quadratic attraction towards the goal,

$$V_g(\mathbf{x}) = \frac{1}{2} \beta \|\mathbf{x} - \mathbf{x}_g\|^2. \quad (10)$$

Obstacle repulsion uses a compactly supported polynomial expressed on the edge-to-edge clearance of Equation (1):

$$V_{o,k}(\mathbf{x}) = \begin{cases} \gamma g_{o,k} \left[1 - \left(\frac{d_k(\mathbf{x})}{R} \right)^2 \right]^p & \text{if } 0 < d_k(\mathbf{x}) < R, \\ 0 & \text{if } d_k(\mathbf{x}) \geq R, \end{cases} \quad (11)$$

where $g_{o,k}$ is an obstacle-specific repulsion gain. It is equal to one for every obstacle in the reported campaign. With $p = 3$, the potential is C^2 at the boundary $d_k = R$ of its support. Two properties of this choice matter. First, the potential is defined on the *physical margin* rather than on the centre-to-centre distance, so the robot radius is embedded in the formulation itself. Second, smoothness is not cosmetic: quasi-Newton minimisation degrades sharply on potentials that are only C^0 at the edge of their support, which is what motivated replacing an earlier inverse-distance barrier.

Arena walls are treated as four half-planes. Writing s for the signed distance to an effective boundary — the geometric boundary offset inwards by r_{rob} plus an arena margin — each wall contributes

$$V_w(s) = \begin{cases} \gamma_w \left(1 - \frac{s}{R_w} \right)^p & \text{if } s < R_w, \\ 0 & \text{if } s \geq R_w, \end{cases} \quad (12)$$

with the same exponent $p = 3$; the value of R_w used for the reported campaign is given in Table 3.

3.3.3 Discretisation

The horizon is discretised on a uniform grid $t_k = k \Delta t$, $k = 0, \dots, N$, with $\Delta t = T/N$. The endpoints are fixed, so the unknowns are the $2(N-1)$ coordinates of the interior points. The action is discretised with the midpoint rule:

$$J_h = \sum_{k=0}^{N-1} \Delta t \left[\frac{\alpha}{2} \left\| \frac{\mathbf{x}_{k+1} - \mathbf{x}_k}{\Delta t} \right\|^2 + V\left(\frac{\mathbf{x}_{k+1} + \mathbf{x}_k}{2}\right) \right] + \lambda_s J_s + J_v + J_a. \quad (13)$$

The smoothing term penalises discrete curvature,

$$J_s = \sum_k \|\mathbf{x}_{k+1} - 2\mathbf{x}_k + \mathbf{x}_{k-1}\|^2, \quad (14)$$

and the two kinematic terms penalise violations of the speed and acceleration bounds,

$$J_v = \lambda_v \sum_k \left[\max \left(0, \frac{\|\mathbf{v}_k\|^2 - v_{\max}^2}{v_{\max}^2} \right) \right]^p, \quad (15)$$

$$J_a = \lambda_a \sum_k \left[\max \left(0, \frac{\|\mathbf{a}_k\|^2 - a_{\max}^2}{a_{\max}^2} \right) \right]^p.$$

These are *soft* penalties, not hard constraints. The nominal trajectory therefore carries no formal guarantee that the kinematic bounds are respected; hard saturation occurs only in the corrector of Section 3.4. This is stated explicitly because it determines what the nominal stage can and cannot be claimed to deliver.

Table 3 lists the weights used to generate the nominal trajectories of the reported campaign. *The same values are used by the real-time runner and by the paired simulations*, so that the two start from the same plan; the consequence is verified in Section 4.4. The weights were tuned empirically and are reported for reproducibility rather than as the outcome of a documented design procedure.

Table 3. Weights of the discrete cost of Equation (13) used to generate the nominal trajectories of the reported campaign.

Term	Symbol	Value
Kinetic weight	α	1.0
Goal attraction	β	0.20
Obstacle repulsion	γ	5.0
Arena penalty	λ_{ar}	8.0
Wall repulsion	γ_w	4.0
Wall influence distance (m)	R_w	0.18
Velocity penalty	λ_v	1.0
Acceleration penalty	λ_a	1.0
Speed bound (m/s)	$v_{\max}$	0.35
Acceleration bound (m/s ²)	$a_{\max}$	0.8
Smoothing	λ_s	0.08
Potential exponent	p	3
Horizon (s)	T	5.0
Intervals	N	30

The full analytic gradient of Equation (13) is derived and supplied to the optimiser, and was validated against centred

finite differences. Its expression is given in Appendix A. The motivation is quantitative: with $2(N - 1)$ design variables, a finite-difference gradient computed by forward differences requires one additional cost evaluation per variable after the reference evaluation, and each cost evaluation sweeps every obstacle at every node.

3.3.4 Nominal generators

Four nominal generators are used in this work. Only the first is used for the experimental campaign; the other three serve as comparison points and ablations.

Variational (L-BFGS-B). Minimises the discrete action (13) with a limited-memory bound-constrained quasi-Newton method [33], using the analytic gradient and the implementation of [34]. A parametric multi-start generates a family of candidates — the straight line plus sinusoidal deviations of alternating sign and increasing amplitude. Candidates that leave the effective arena are rejected. Obstacle clearance is recorded but is not a rejection criterion: penetrations receive the finite core penalty of Equation (A.2). The lowest-cost remaining candidate is used as a warm start. The optimisation is *local*: the result depends on the initialisation and on the potential weights.

Euler–Lagrange residual. Solves the discrete stationarity condition directly by driving the residual

$$\mathbf{R}_k = \alpha \frac{\mathbf{x}_{k+1} - 2\mathbf{x}_k + \mathbf{x}_{k-1}}{\Delta t^2} - \nabla V_{\text{EL}}(\mathbf{x}_k) = \mathbf{0} \quad (16)$$

to zero at every interior node with a hybrid root-finding method [35]. This generator uses dedicated smoothed potentials V_{EL} , built on a regularised distance $d_\varepsilon = \sqrt{\|\mathbf{x} - \mathbf{c}\|^2 + \varepsilon^2}$ and a compact bump. Those potentials are not term-by-term identical to those of Equation (13), and the two generators must therefore not be presented as two numerical methods applied to a strictly identical function.

Straight line. Returns the affine interpolation

$$\mathbf{x}_{\text{ref}}(t) = \mathbf{x}_0 + \frac{t}{T} (\mathbf{x}_g - \mathbf{x}_0), \quad (17)$$

deliberately ignoring obstacles and potentials. This is an ablation: if the straight line combined with the corrector performs well, then a substantial part of the observed performance comes from the local correction rather than from the global plan.

No-plan ablation. Produces no nominal trajectory at all. The attraction term of the corrector then points directly at the goal, as given in Equation (19) below. This is the minimal architecture, with zero offline computation.

Table 4 summarises the four generators, and Figure 6 illustrates the passage from the candidate family to the retained nominal trajectory and then to the corrected motion.

3.4 Causal local correction

3.4.1 Force model

The corrector runs at every control step of the real-time loop. It treats the robot as a point of mass m subject to a total force

$$\mathbf{F}_k = \mathbf{F}_k^{\text{att}} + \mathbf{F}_k^{\text{mob}} + \mathbf{F}_k^{\text{stat}} + \mathbf{F}_k^{\text{bnd}} - c_d \mathbf{v}_k, \quad (18)$$

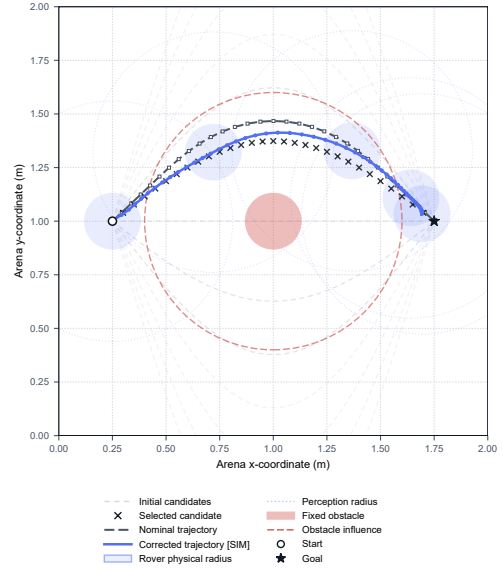


Figure 6. From the parametric candidate family to the retained nominal trajectory, and from the nominal trajectory to the causally corrected motion.

where the last term is viscous damping. The attraction pulls the robot towards a point on the nominal trajectory taken *ahead* of the current time,

$$\mathbf{F}_k^{\text{att}} = k_a [\mathbf{x}_{\text{ref}}(t_k + t_\ell) - \mathbf{x}_k], \quad (19)$$

with t_ℓ the look-ahead time. Reading the *nominal* trajectory ahead of time is not a violation of causality: $\mathbf{x}_{\text{ref}}$ is a fixed function computed offline from the static scene, and evaluating it at $t_k + t_\ell$ queries a plan, not the future state of the world. For the no-plan ablation, $\mathbf{x}_{\text{ref}}(t_k + t_\ell)$ is replaced by $\mathbf{x}_g$.

The three repulsive terms share a common kernel, expressed on the edge-to-edge clearance d of Equation (1) and active only inside the influence radius H of the corresponding element:

$$\mathbf{F}^{\text{rep}} = g \max\left(0, \frac{1}{d_\varepsilon^2} - \frac{1}{H^2}\right) \hat{\mathbf{n}}, \quad d_\varepsilon = \max(d, \varepsilon), \quad (20)$$

where $\hat{\mathbf{n}}$ is the unit vector pointing away from the element, g is the corresponding gain (k_m , k_s or k_b) and ε is a regularisation floor. Subtracting $1/H^2$ makes the force vanish continuously at the edge of the influence region rather than jumping to zero.

The three influence radii differ by an order of magnitude, consistent with the role each repulsion plays. Moving agents are absent from the nominal scene by construction, so the corrector alone is responsible for avoiding them and must react from a distance: $H = 0.60$ m, equal to the perception radius. Fixed obstacles are already routed around by the nominal trajectory, so the static term acts as a last-resort safeguard behind that plan: $H = 0.07$ m on the clearance. The arena boundary uses $H = 0.20$ m. This asymmetry also explains why the smallest clearances observed against fixed obstacles are tighter than those against moving agents (Section 4.3.5).

All three values are expressed on the edge-to-edge clearance of Equation (1) and therefore depend on the robot radius; they correspond to $r_{\text{rob}} = 0.130$ m as used throughout the reported campaign.

The acceleration and velocity are then saturated, in that order:

$$\mathbf{a}_k = \text{clip}_{a_{\text{max}}} \left(\frac{\mathbf{F}_k}{m} \right), \quad \mathbf{v}_{k+1} = \text{clip}_{v_{\text{max}}} (\mathbf{v}_k + \Delta t \mathbf{a}_k), \quad (21)$$

Table 4. Nominal trajectory generators.

Generator	Principle	Interest	Main limitation
Variational (L-BFGS-B)	Minimises the discrete action	Rich global plan, obstacles embedded in the cost	Longer computation, dependence on initialisation
Euler–Lagrange residual	Cancels the discrete stationarity residual	Direct mechanical reading; residual usable as a diagnostic	Sensitive convergence; potentials not identical to the variational cost
Straight line	Affine interpolation from start to goal	Very fast and fully interpretable baseline	The nominal path may cross obstacles
No-plan ablation	No nominal at all; direct attraction to the goal	Minimal architecture, zero offline cost	Local minima; safe stalling away from the goal

where $\text{clip}_c(\mathbf{u})$ rescales $\mathbf{u}$ to norm c when $\|\mathbf{u}\| > c$ and leaves it unchanged otherwise. Unlike the soft penalties of Equation (15), these hard bounds are satisfied at every step.

3.4.2 Integration schemes

Three integration schemes are used in this work. The corrector used for the experimental campaign is the first.

Semi-implicit Euler corrector. One force evaluation per step, with the position updated using the *already updated* velocity:

$$\begin{aligned} \mathbf{a}_k &= \frac{\mathbf{F}(\mathbf{x}_k, \mathbf{v}_k, t_k)}{m}, \\ \mathbf{v}_{k+1} &= \text{clip}_{v_{\max}}(\mathbf{v}_k + \Delta t \mathbf{a}_k), \\ \mathbf{x}_{k+1} &= \mathbf{x}_k + \Delta t \mathbf{v}_{k+1}. \end{aligned} \quad (22)$$

Velocity-Verlet corrector. A predictor–corrector scheme in the sense of [36], with two force evaluations per step:

$$\begin{aligned} \mathbf{a}_k &= \frac{\mathbf{F}(\mathbf{x}_k, \mathbf{v}_k, t_k)}{m}, \\ \mathbf{x}^{\text{pred}} &= \mathbf{x}_k + \Delta t \mathbf{v}_k + \frac{1}{2} \Delta t^2 \mathbf{a}_k, \\ \mathbf{v}^{1/2} &= \text{clip}_{v_{\max}}\left(\mathbf{v}_k + \frac{1}{2} \Delta t \mathbf{a}_k\right), \\ \mathbf{a}^{\text{pred}} &= \frac{\mathbf{F}(\mathbf{x}^{\text{pred}}, \mathbf{v}^{1/2}, t_k)}{m}, \\ \mathbf{v}_{k+1} &= \text{clip}_{v_{\max}}\left[\mathbf{v}_k + \frac{1}{2} \Delta t (\mathbf{a}_k + \mathbf{a}^{\text{pred}})\right], \\ \mathbf{x}_{k+1} &= \mathbf{x}^{\text{pred}}. \end{aligned} \quad (23)$$

The predicted acceleration in Equation (23) is evaluated at the *same* instant t_k , and this is deliberate. The scheme predicts the position of the *robot*; it never predicts the future state of the obstacles. Evaluating $\mathbf{F}$ at t_{k+1} would require the agent positions at t_{k+1} and would break causality, which is precisely the failure mode this architecture is designed to exclude.

Stiff-repulsion corrector. Identical semi-implicit integration, but with a Lennard-Jones type repulsion [37] replacing the kernel of Equation (20):

$$V_J(d) = \frac{C}{d_{\varepsilon}^{12}}, \quad F_J(d) = \frac{12C}{d_{\varepsilon}^{13}}, \quad d_{\varepsilon} = \max(d, \varepsilon), \quad (24)$$

active only within the influence radius. The d^{-13} growth produces an extremely strong reaction near contact. Because

the acceleration is bounded by Equation (21), an excessive gain does not translate into an arbitrarily large response but keeps the corrector saturated, which degrades progress towards the goal.

Table 5 summarises the three schemes.

Figure 7 shows the decomposition of Equation (18) along a measured run.

3.4.3 How causality is enforced

Causality as defined in Section 2.3 is guaranteed by construction rather than by discipline. Four mechanisms enforce it:

- the world description passed to the nominal optimiser is built by a dedicated routine that *excludes moving agents*, so that the offline stage cannot see them at any instant, past or future;
- the interface of a moving agent exposes its position only at the current time, so that no code path can query an agent position at a later instant;
- the corrector observes only the agents lying inside a finite perception radius r_p around the robot, so that its input is a local and current view of the scene;
- in the predictor–corrector scheme, the second force evaluation uses a predicted *robot* position at the *current* time, as noted after Equation (23).

Table 6 lists the corrector parameters used for the experimental campaign. A single coefficient set was used for every run of the campaign, with no retuning between runs.

3.5 Wheel-level command generation

The corrector of Section 3.4 produces a force in the plane. The robot accepts neither a force nor a velocity: it accepts two scalar wheel commands. This subsection describes the conversion, and the open-loop characterisation of the drivetrain that dictates its structure.

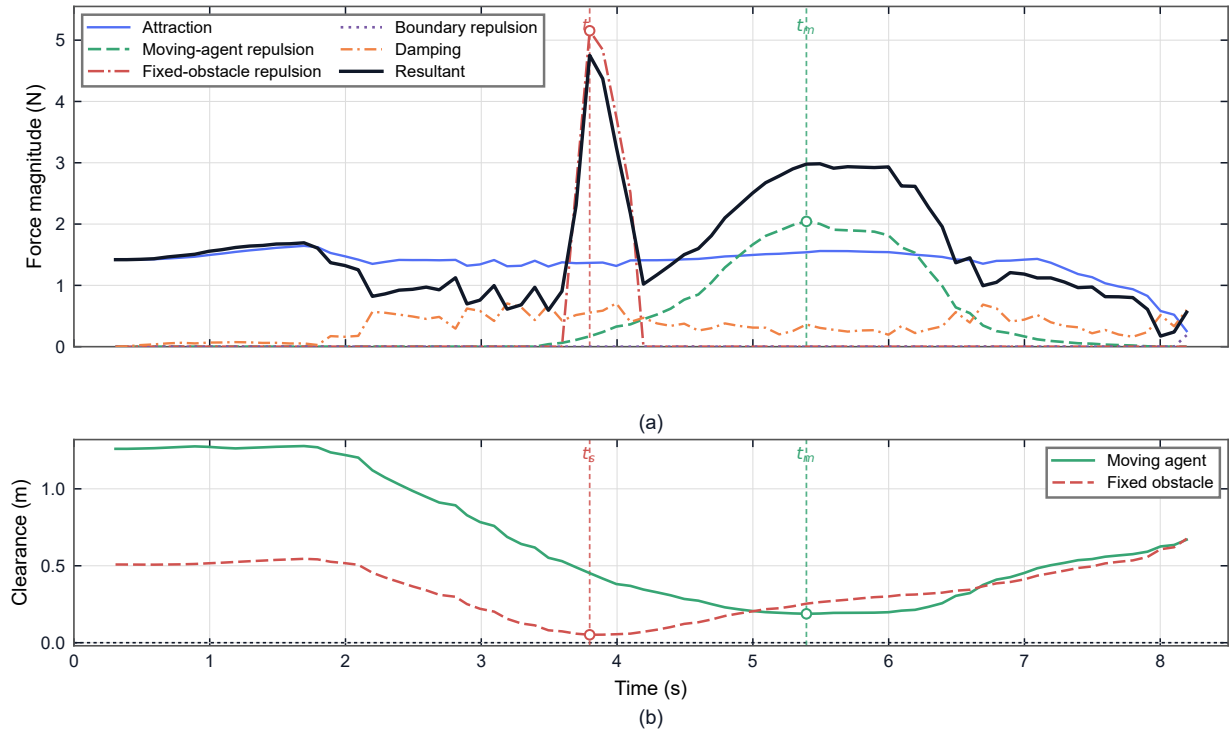
3.5.1 From force to wheel commands

The conversion proceeds in three stages. First the corrector force is integrated into a velocity, exactly as in Equation (22), with the acceleration clipped in norm to $a_{\max}$ and the resulting velocity clipped to $v_{\max}$. The direction of that velocity defines the desired heading ϕ , and the heading error with respect to the measured robot orientation is

$$\theta = \text{wrap}(\phi - \psi) \in [-\pi, \pi), \quad (25)$$

Table 5. Causal local correctors.

Corrector	Structural difference	Interest	Main limitation
Semi-implicit Euler	One force evaluation per step, standard repulsion kernel	Simple, lowest computational cost	Sensitivity to step size and gains
Velocity-Verlet	Position predicted, then force re-evaluated at the same instant	Better balanced integration with identical forces	Two force evaluations per step
Stiff repulsion	Repulsion in d^{-13}	Very strong reaction close to contact	Stiffness, saturation, poor terminal progress

**Figure 7.** Decomposition of the total corrector force along a measured run: attraction, moving-agent repulsion, static repulsion, boundary repulsion and damping.**Table 6.** Causal corrector parameters used for the experimental campaign. The same coefficient set was applied to every run, with no retuning. The mass m is the measured mass of the controlled robot, weighed in its run configuration (Table 1).

Parameter	Symbol	Value
Attraction gain	k_a	5.0
Damping coefficient	c_d	1.4
Moving-agent repulsion gain	k_m	0.08
Static repulsion gain	k_s	0.03
Boundary repulsion gain	k_b	0.02
Perception radius	r_p	0.60 m
Look-ahead time	t_ℓ	0.8 s
Boundary influence distance	R_w	0.20 m
Robot mass (measured)	m	1.2588 kg
Speed bound	$v_{\max}$	0.25 m/s
Acceleration bound	$a_{\max}$	1.0 m/s ²
Regularisation floor	ε	1×10^{-4} m

where ψ is the marker orientation corrected by the fixed forward-axis offset of Section 3.2.5.

Second, a yaw-rate request is formed from a feed-forward and a feedback term,

$$\dot{\psi}_{\text{req}} = \text{sat}(\kappa_{\text{ff}} \dot{\phi}, k_\omega) + 2k_\theta k_\omega \theta, \quad \Delta u = \frac{\dot{\psi}_{\text{req}}}{k_\omega}, \quad (26)$$

in which $\dot{\phi}$ is the rate of change of the desired heading between two control steps, κ_{ff} weights that anticipation, k_θ is a heading gain and k_ω is the angular gain of the inverse motor model. Here $\text{sat}(z; c) = \min\{\max\{z, -c\}, c\}$ is scalar saturation, applied only to the feed-forward term. The resulting differential request Δu is converted into the wheel commands below.

Third, the controller selects between turning in place and driving. The selection uses hysteresis, with distinct entry and exit thresholds $\theta_{\text{in}} > \theta_{\text{out}}$, so that a heading error oscillating about a single threshold cannot make the robot alternate between

the two regimes at every control step:

$$\begin{aligned} \text{turning: } u_v &= 0, \\ u_\omega &= \text{sat}\left(\frac{1}{2}\Delta u, u_\omega^{\max}\right), \quad |u_\omega| \geq u_\omega^{\min}, \\ \text{driving: } u_v &= \text{sgn}(v) \max\left(u_v^{\min}, \min\left(u_v^{\max}, \frac{|v|}{k_v}\right)\right), \\ |u_\omega| &\leq \eta |u_v|. \end{aligned} \quad (27)$$

where k_v is the linear gain of the inverse motor model. The two wheel commands follow as

$$u_L = u_v - u_\omega, \quad u_R = u_v + u_\omega, \quad (28)$$

jointly rescaled if either exceeds the regime limit, so that their ratio — and therefore the commanded curvature — is preserved rather than clipped. A minimum effective value is then applied to any non-zero wheel command, and the pair is smoothed exponentially as given in Equation (A.8) of Appendix A. Table 7 lists the parameters, which are recorded in the log of every run.

Table 7. Wheel-level parameters of the reported campaign, as recorded in the run logs. Commands are dimensionless and normalised to the full scale of the motor driver.

Parameter	Symbol	Value
Linear gain (m/s per command)	k_v	0.943
Angular gain (rad/s per Δu)	k_ω	0.806
Heading gain	k_θ	0.55
Feed-forward weight	κ_{ff}	0.60
Turn-in-place entry threshold (rad)	θ_{in}	1.20
Turn-in-place exit threshold (rad)	θ_{out}	0.65
Maximum driving command	$u_v^{\max}$	0.50
Minimum driving command	$u_v^{\min}$	0.10
Maximum turning command	$u_\omega^{\max}$	0.50
Minimum turning command	$u_\omega^{\min}$	0.30
Minimum effective wheel command	$u_{\text{eff}}^{\min}$	0.05
Maximum arc turn ratio	η	1.00
Command smoothing factor	α_{cmd}	0.60
Control period (s)	Δt_c	0.05

3.5.2 Reverse motion

A differential-drive robot whose target lies behind it can either rotate by half a turn or drive backwards. The controller used in the reported campaign takes the second option: when the heading error exceeds an entry threshold, the sign of the commanded speed is inverted and the desired heading is reflected by π , so that the rear of the chassis becomes the leading face. The regime is left through a lower exit threshold, again with hysteresis, and the smoothing history is cleared at each transition so that a command of the previous direction cannot survive into the new one.

Both blocks of the reported campaign use the same thresholds, entry at 92.01° and exit at 91.97° , so that the hysteresis band is 0.04° at the displayed precision and therefore effectively closed. Reverse motion was triggered in every run of both blocks, for a median of 12.79 % and 10.00 % of the control samples respectively.

Because no run of either block was performed with the mechanism disabled, its individual contribution cannot be isolated from these data, and no such claim is made.

3.5.3 Open-loop characterisation of the drivetrain

Because the drivetrain has no wheel encoders, the mapping from command to realised velocity is an open-loop property of the hardware and was measured directly. Two campaigns were run on the instrumented arena, with the overhead camera as the only measurement instrument. Both were run on the controlled robot in the configuration of the reported runs — same onboard computer, power supply and fixtures, and therefore the same mass — so that the identified mapping describes the platform that was actually used.

Translation. Thirty trials covered six command levels between 0.08 and 0.25, five repetitions each, with complete marker detection. The relation between command c and realised speed v is *affine, not proportional*:

$$v = 1.5903 c - 0.0888 \quad (R^2 = 0.9959), \quad (29)$$

against $R^2 = 0.887$ for the best fit through the origin. The coefficient of determination is reported over the thirty individual trials; the same fit taken over the six level means gives 0.9991 for identical slope and intercept. The affine fit predicts zero velocity at $c_0 = 0.0558$. Since the experiments only covered commands from $c = 0.08$ upwards, this value is an extrapolated zero-motion point rather than a measured dead-zone threshold.

The repeatability of the platform follows the same pattern. The relative standard deviation over five repetitions falls from 28.0 % at $c = 0.08$, the lowest tested command, to 3.4 % at $c = 0.14$ and 1.1 % at $c = 0.25$. Thus, over the tested range, repeatability improves as the command increases. These observations motivate avoiding very low driving commands, but do not identify a static threshold below which motion ceases. As discussed below, the minimum driving command $u_v^{\min}$ of Equation (27) remains an empirically tuned controller parameter.

Parasitic motion in translation is small. The mean absolute lateral drift is 1.03 mm with a maximum of 4.21 mm, and the parasitic yaw stays below 2.88° over advances of 1.8–15.6 cm. In straight-line motion the chassis is therefore close to a symmetric differential drive.

Rotation. Seventy trials covered seven command levels between 0.25 and 0.45, in both directions, five repetitions each. The affine character is far more pronounced than in translation: a fit through the origin reaches only $R^2 = 0.374$ and is unusable, whereas the affine fit gives $R^2 = 0.869$ and predicts zero yaw rate at $c_0 = 0.268$. This is a zero-motion point estimated from the fit, not a measured dead-zone threshold: a non-zero yaw rate was already observed at the lowest tested command, $c = 0.25$. The experiment therefore demonstrates an affine rotational response but does not establish a dead zone.

The departure from proportionality means that a fixed linear inverse is least accurate in the low-command regime. The controller handles that regime through the switching thresholds θ_{in} and θ_{out} and the floor $u_\omega^{\min}$. The campaign does not, however, identify a minimum command that produces rotation; these quantities are empirically tuned software parameters rather than measured hardware thresholds.

Left–right asymmetry. Rotation to the right was faster than rotation to the left at all seven command levels tested. The

median ratio of realised yaw is 1.187, ranging from 1.047 to 1.383, and the ratio of the fitted slopes is 1.084; the difference between the two figures comes from the different fitted offsets in the two directions. The rotation is also not a perfect pivot: the centre of the robot drifts by 3.19 mm on average and by up to 8.51 mm, the drift growing with the command level.

This asymmetry is not compensated: Equation (28) is symmetric in the two wheels, and the inverse model uses a single angular gain for both directions. It is carried forward as an identified source of deviation between plan and execution in Section 5.3. The centre drift, at 8.5 mm in the worst case, is more than an order of magnitude below the safety radius r_{rob} and is therefore absorbed by the conservative bounding disc.

3.5.4 Status of the inverse model

The inverse model of Equations (26) and (27) is a fixed linear gain in each axis. It is *deliberately simpler than the affine relation measured above*, and its gains are not those of Equation (29). The project records do not document how the values of Table 7 were obtained. They are therefore reported as *empirically tuned controller parameters*, rather than as parameters identified from the campaigns of this subsection. Those campaigns characterise the platform; they do not calibrate the controller.

This mismatch has a concrete effect. A linear gain smaller than the measured affine slope makes the inversion over-command slightly and moves the operating point away from the low-command region, where the mismatch with the affine response is greatest. The records do not establish that this was the intent, and it is not presented as such.

Figure 8 reports both characterisation campaigns.

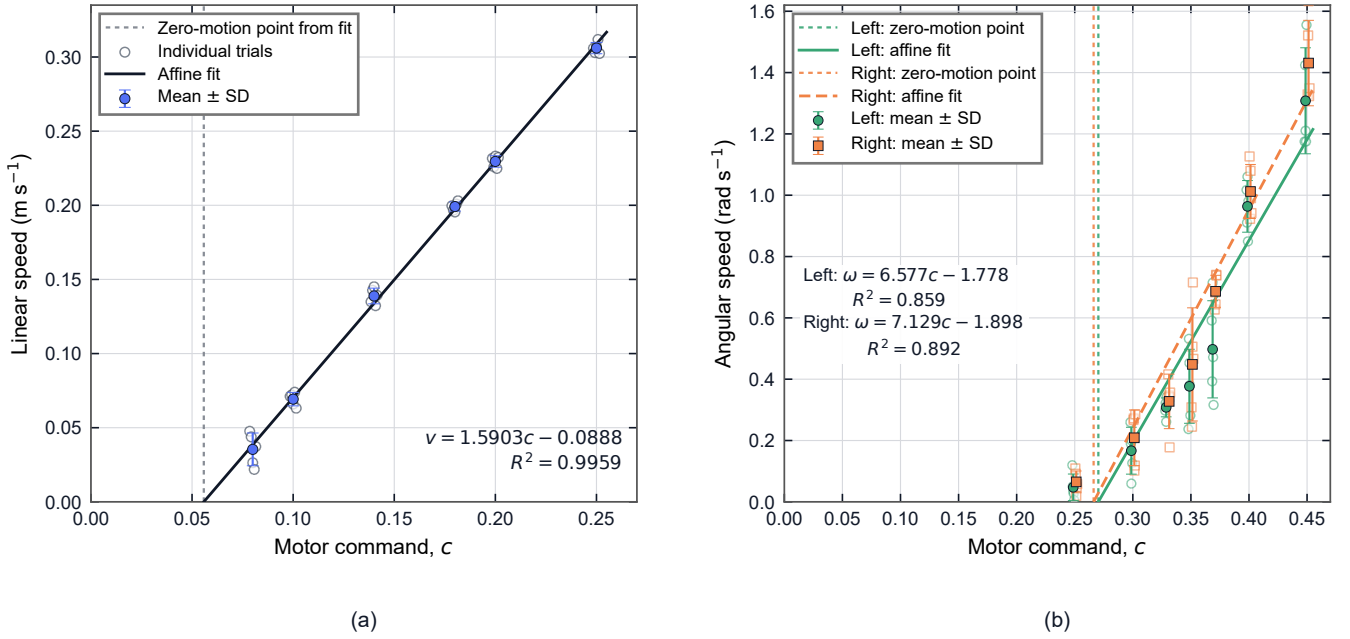


Figure 8. Open-loop characterisation of the drivetrain. Left: realised speed against translation command over thirty trials, with the affine fit and the extrapolated zero-motion point. Right: realised yaw rate against rotation command over seventy trials, separated by direction, with the affine fits and the zero-motion points estimated from them. Error bars span one standard deviation over five repetitions.

3.6 Automatic calibration protocol

The generators of Section 3.3 and the correctors of Section 3.4 can be combined in twelve ways. Comparing them requires that each be tuned on its own terms, since a corrector evaluated with poorly chosen gains says nothing about the corrector. This subsection describes the protocol used to tune each combination independently before comparison. Independence of tuning is not the same as equality of budget, and the sampling budgets actually granted are stated in Section 4.1. It is carried out entirely in simulation.

3.6.1 Design space

Seven parameters of the causal corrector are tuned for every combination. Their initial ranges are listed in Table 8; no gain is ever inherited from another combination.

Table 8. Tuned parameters and their initial ranges. Gains are dimensionless in the numerical model of Section 3.4.

Parameter	Symbol	Lower	Upper
Attraction gain	k_a	1.50	6.50
Damping coefficient	c_d	2.00	8.00
Moving-agent repulsion gain	k_m	0.020	0.090
Static repulsion gain	k_s	0.015	0.050
Boundary repulsion gain	k_b	0.010	0.035
Perception radius (m)	r_p	0.45	0.95
Look-ahead time (s)	t_ℓ	0.45	1.05

3.6.2 Scenario bank

Twenty-six planar scenarios are used, differing in the number and arrangement of fixed obstacles and moving agents. They fall into four descriptive families — corridors and barriers, scattered obstacle fields, crossing traffic, and an unobstructed reference — with obstacle counts ranging from none to forty-five. All share a horizon of 5 s, and all but two use thirty nodes.

Sixteen of these scenarios form the calibration set and the

remaining ten are held out. *The split was fixed a priori and the held-out scenarios were not selected on difficulty*; they are simply the complement of the calibration set. Held out means not used for tuning: the retained profiles are afterwards evaluated on all twenty-six.

3.6.3 Search procedure

The search proceeds in three stages of Latin hypercube sampling [38] over the parameter box. After each stage the box is contracted around the surviving candidates: the best fraction of profiles is retained, the tenth and ninetieth percentiles of each parameter are taken over that subset, and the new bounds are those percentiles widened by a fixed fraction of their spread and clipped to the original range. The contraction fraction is 0.25 at the second stage and 0.18 at the third.

Sampling draws 400, 640 and 400 profiles per stage under the standard budget, and 900, 1400 and 900 under the extended one; the first stage additionally evaluates the uncontracted reference profile. The three stages use one seed and two fixed offsets, so that a single integer determines the whole design.

3.6.4 Objective

Each profile is run on every non-excluded scenario and scored by a single quantity, *minimised*:

$$S = \Lambda (n_m + n_s + n_b) + 100 \sum_i w_i \Pi_i, \quad (30)$$

where n_m , n_s and n_b count the runs ending in a moving-agent collision, a fixed-obstacle collision and an arena exit, and $\Lambda = 6000$ is a hard penalty per such event. The seven weighted terms Π_i measure the minimum clearance to the moving agents ($w = 0.26$), to the fixed obstacles (0.20) and to the arena boundary (0.14), the mean final distance to the goal (0.18), the mean acceleration saturation rate (0.12), a logarithmic measure of the commanded force (0.06) and the mean tracking error with respect to the nominal (0.04). Their full expressions are given in Appendix A.4.

Two properties of Equation (30) matter for the interpretation of the results. Each term is scaled by a *fixed* threshold rather than normalised over the population of profiles, so the score of a profile does not depend on the other profiles. And the weighted block is bounded above by 406, against 6000 for a single hard event, so any run ending in a collision or an arena exit makes its profile rank worse than every collision-free one.

3.6.5 Scenario quarantine

A scenario on which a combination fails almost everywhere carries no information about that combination's gains, and its hard penalties would saturate the score. After the first stage, any scenario whose collision-free rate falls below 15 % across the profiles of the current combination is therefore placed in quarantine. It is reinstated after the second stage if that rate reaches 50 %, and after the third if it reaches 80 %.

A *quarantined scenario is removed from the score rather than penalised in it*. The consequence is stated here because it governs how Section 4.1 must be read: the twelve combinations are not scored over a common set of scenarios, and the size of each scoring set is reported alongside the results.

3.6.6 Volume

The complete protocol comprises 367 374 planned runs across the twelve combinations, for a cumulative computation time of 77.5 h.

3.7 Metrics and evaluation criteria

3.7.1 Run-level indicators

Every run is summarised by the three binary indicators introduced in Equations (2) to (4). Over a set $\mathcal{R}$ of n runs, the corresponding rates are

$$\begin{aligned} R_s &= \frac{100}{n} \sum_{r \in \mathcal{R}} I_s(r), \\ R_g &= \frac{100}{n} \sum_{r \in \mathcal{R}} I_g(r; \tau), \\ R_j &= \frac{100}{n} \sum_{r \in \mathcal{R}} I_j(r; \tau). \end{aligned} \quad (31)$$

expressed in per cent and satisfying $R_j \leq \min(R_s, R_g)$. Outcomes are further partitioned into four mutually exclusive categories summing to 100 %: complete success ($I_j = 1$), safe but goal missed ($I_s = 1, I_g = 0$), unsafe without execution error, and execution error. The second category is the one that matters most in practice, because it isolates the failure mode in which a controller protects itself into immobility.

These quantities are *frequencies over a finite set of runs*, not probabilities. They carry no confidence interval, and they are reported as such throughout the paper.

3.7.2 Clearance

Clearance is measured edge-to-edge between the conservative bounding discs used by the controller (radius $r_{\text{rob}} = 0.130$ m per robot), not between the physical chassis outlines. A negative value therefore means that the two safety discs overlapped, *not* that the robots made contact. This distinction is essential to the interpretation of Section 4.3 and is repeated wherever a negative clearance is reported. Two clearances are logged at every control step: the minimum over the observed moving agents, and the minimum over the fixed obstacles.

3.7.3 Terminal condition

A run is declared successful when the measured distance to the goal falls below a fixed tolerance,

$$\|\mathbf{x}(t) - \mathbf{x}_g\| \leq \tau, \quad \tau = 0.08 \text{ m}, \quad (32)$$

evaluated at every control step. The consequence must be stated explicitly: the final distance recorded for a successful run indicates *where the stopping condition was triggered*, not the accuracy the controller could ultimately reach. At 0.25 m/s and 20 Hz the robot advances about 1.25 cm per control step, which bounds the achievable spread of that quantity. Final distances are therefore never interpreted as a positioning accuracy in this paper.

Two further stopping conditions terminate a run without reaching the goal: a maximum duration, and a safety stop triggered when any selected moving-agent marker remains undetected beyond the retention interval.

3.7.4 Classification of moving agents

A marker being *observed* does not imply that the corresponding agent *moved*. Because the agents were placed and commanded manually (Section 3.1.4), some runs contain a detected agent that remained essentially stationary, and in those runs no avoidance was demanded of the controlled robot. Reporting such runs together with genuine avoidance runs would overstate the difficulty of the campaign.

An observed agent is therefore counted as *moving* when its net displacement over the run exceeds

$$\delta_{\min} = 0.30 \text{ m}. \quad (33)$$

This threshold is not critical: across the campaign, the net displacements form two clearly separated populations, one below 0.12 m and one above 0.39 m, with no agent in between. Results in Section 4.3 are stratified on this criterion.

3.7.5 Simulation-versus-experiment agreement

Each physical run is paired with a simulation of the same scenario — same start, same goal, same fixed obstacles, same coefficients and *the same kinematic bounds* — driven by the recorded trajectories of the moving agents. Agreement is quantified geometrically. Writing $d(\mathbf{p}, \Gamma)$ for the Euclidean distance from a point $\mathbf{p}$ to a polyline Γ , and Γ_m, Γ_s for the measured and simulated paths, the reported quantities are the mean and maximum point-to-path departure

$$\bar{e} = \frac{1}{|\Gamma_m|} \sum_{\mathbf{p} \in \Gamma_m} d(\mathbf{p}, \Gamma_s), \quad e_{\max} = \max_{\mathbf{p} \in \Gamma_m} d(\mathbf{p}, \Gamma_s), \quad (34)$$

together with the symmetric Hausdorff distance

$$d_H = \max \left\{ e_{\max}, \max_{\mathbf{q} \in \Gamma_s} d(\mathbf{q}, \Gamma_m) \right\},$$

and the ratio of executed to simulated path length.

The measure in Equation (34) ignores any time shift between the two trajectories and penalises only geometric departure. This geometric comparison deliberately ignores timing mismatch.

4 Results

4.1 Solver-controller comparison in simulation

This subsection reports the twelve solver-controller combinations tuned by the protocol of Section 3.6. It is a simulation study, independent of the experimental campaign of Sections 4.2 to 4.4.

4.1.1 Reading the comparison

Three rates are reported for each combination, all defined in Section 3.7: the safety rate R_s , the goal rate R_g at a terminal tolerance of 10 cm, and the joint rate R_j . They are given both over the whole final sampling region, which measures how wide a set of gains works, and over the best-ranked profiles of each combination — the four highest-ranked, the same number for every combination — which measures how well the retained tuning performs.

Two qualifications apply throughout and are stated once. First, the quarantine of Section 3.6 removes scenarios from a combination's score rather than penalising them, so *the combinations are not scored over a common set of scenarios*; the size of each scoring set is given in Table 9 and ranges from six to fourteen of the sixteen calibration scenarios. Second, each combination is evaluated once, without repetition over independent seeds, so the reported percentages are frequencies over a finite set of profiles and scenarios. Any statement that one combination is best is therefore a statement about the subset of scenarios on which it was scored. For the same reason, no interval accompanies any rate reported in this subsection or drawn in the figures that support it; the absence is deliberate.

4.1.2 Safety alone is a misleading indicator

The most instructive rows of Table 9 are the last three. *The architecture with no global plan at all achieves the highest safety rate of the study* — 96.15 % of runs without collision for two of its three correctors — *together with the lowest joint rate*, at 3.85 %, and a joint rate of zero on the held-out scenarios.

This is not a safety failure but a progression failure. Deprived of a global plan, the attraction term points straight at the goal, and the robot comes to rest wherever that attraction balances the repulsions. It stops safely, and it stops far away. Reporting the safety rate alone would rank this ablation first.

The observation is the empirical counterpart of the argument made in Section 2.2, and it is the reason the joint indicator is used throughout this work.

4.1.3 Two families of failure

The combinations fail in two distinguishable ways, and the pair of rates identifies which.

A high safety rate with a low goal rate indicates the progression failure just described, and it characterises the no-plan ablation under every corrector. A low safety rate indicates the opposite mechanism: the stiff-repulsion corrector keeps the actuator saturated near obstacles, and the resulting motion violates the very margin the repulsion was meant to protect. The combination of the variational generator with that corrector falls to 37.23 % safety over the region — *below the ablation that has no plan at all*. A good global plan does not compensate for a local law that is too stiff.

4.1.4 The value of the global plan

The straight-line baseline, which ignores obstacles entirely at the planning stage, ranks third over the best profiles at 48.08 %. That a plan which crosses obstacles still supports a joint rate of that order indicates that a substantial part of the performance comes from the local correction rather than from the global plan.

A gap nevertheless separates it from the variational generator, at 65.38 % and 63.46 % for the two best combinations, and those two also generalise best, both reaching 70 % on the held-

out scenarios. Figure 12 places every combination on the calibration-held-out plane and shows that this is not a single favourable pairing: the two variational combinations are the only ones that sit high on both axes at once. *These are success frequencies, and they do not apportion the avoidance between the two layers*. The reading carried forward in Section 5.2 — that the global plan removes the need for the corrector to resolve a global detour through local forces — is an interpretation consistent with them, not a measurement of it.

Among the correctors, the Velocity-Verlet scheme performs better than semi-implicit Euler when paired with the Euler-Lagrange generator, at 42.31 % against 31.73 %. A predictor-corrector integration tolerating a less regular reference would produce that ordering, but the campaign does not measure the regularity of the nominal and so does not establish the mechanism.

4.1.5 The ranking depends on the tolerance

Figures 9 to 13 report the study. Figure 9 gives the rates over the whole sampling region and Figure 10 over the best-ranked profiles, which is the pair of views summarised in Table 9; Figure 11 recasts the same runs as an exclusive decomposition, so that the two failure families are read directly rather than inferred from a difference of rates. Figure 13 is the most consequential of the five: it sweeps the terminal tolerance from zero to 100 cm and shows that *the ordering of the combinations changes with the precision demanded*. At the 10 cm reference used above, the variational generator with the Velocity-Verlet corrector leads. At 100 cm it no longer does: the straight-line generator with the semi-implicit Euler corrector reaches 73.08 % there, the same value as the best variational combination, and the two are indistinguishable over the last third of the sweep. *The advantage the variational plan holds at a tight tolerance is an advantage in reaching the goal precisely, and it disappears once arriving anywhere near it is enough*.

The shape of each curve is itself a diagnostic. A curve that rises slowly as the tolerance is relaxed indicates a progression problem, the robot having stopped far from the goal; a curve already high at a tight tolerance but plateauing low indicates a safety problem. *The question of which solver is best is therefore ill-posed until the required terminal precision is stated*.

4.1.6 Scope of the comparison

Three restrictions bound what Table 9 establishes, in addition to the two stated at the outset.

The final sampling region was evaluated on the sixteen calibration scenarios only; the best-ranked profiles were evaluated on all twenty-six. The variable scoring perimeter declared above concerns the tuning stage, which is where the quarantine acts and where the ranking is decided; once four profiles per combination have been retained, they are re-evaluated on the full set, so the best-profile columns of Table 9 do rest on a common basis. What the variable perimeter compromises is therefore the *selection* of the retained profiles, not their subsequent comparison. Sampling budgets are also unequal: the three combinations built on the straight-line generator explored 900 profiles in the final region against 400 for the others, although all twelve retain four. And the two nominal generators do not minimise a strictly identical function, as noted in Section 3.3, so they must not be read as two numerical methods applied to the same objective.

Finally, these results are obtained in simulation, under the

Table 9. The twelve combinations, ordered by joint rate over the best-ranked profiles. Triples are safety / goal / joint rate in percent at a terminal tolerance of 10 cm. The last column gives the number of calibration scenarios that entered the score, out of sixteen.

Nominal	Corrector	Region	Best profiles	R_j cal.	R_j h.o.	Scored
Variational	Velocity-Verlet	72.44 / 57.61 / 48.30	76.92 / 69.23 / 65.38	62.50	70.00	12
Variational	Semi-implicit Euler	72.62 / 64.94 / 50.84	80.77 / 72.12 / 63.46	59.38	70.00	13
Straight line	Semi-implicit Euler	77.89 / 49.30 / 39.38	88.46 / 52.88 / 48.08	46.88	50.00	13
Euler-Lagrange	Velocity-Verlet	62.50 / 50.00 / 37.50	65.38 / 55.77 / 42.31	37.50	50.00	10
Euler-Lagrange	Semi-implicit Euler	63.83 / 44.72 / 30.25	68.27 / 41.35 / 31.73	23.44	45.00	11
Straight line	Velocity-Verlet	77.95 / 33.69 / 29.95	84.62 / 30.77 / 29.81	20.31	45.00	12
Straight line	Stiff-repulsion	77.19 / 18.65 / 18.51	74.04 / 22.12 / 20.19	18.75	22.50	12
Variational	Stiff-repulsion	37.23 / 42.33 / 11.30	49.04 / 41.35 / 18.27	10.94	30.00	6
Euler-Lagrange	Stiff-repulsion	43.27 / 33.66 / 5.33	53.85 / 37.50 / 16.35	6.25	32.50	8
No-plan ablation	Stiff-repulsion	89.69 / 6.19 / 5.97	90.38 / 9.62 / 9.62	6.25	15.00	14
No-plan ablation	Semi-implicit Euler	85.80 / 22.58 / 16.38	96.15 / 7.69 / 3.85	6.25	0.00	13
No-plan ablation	Velocity-Verlet	85.14 / 21.83 / 15.56	96.15 / 7.69 / 3.85	6.25	0.00	13

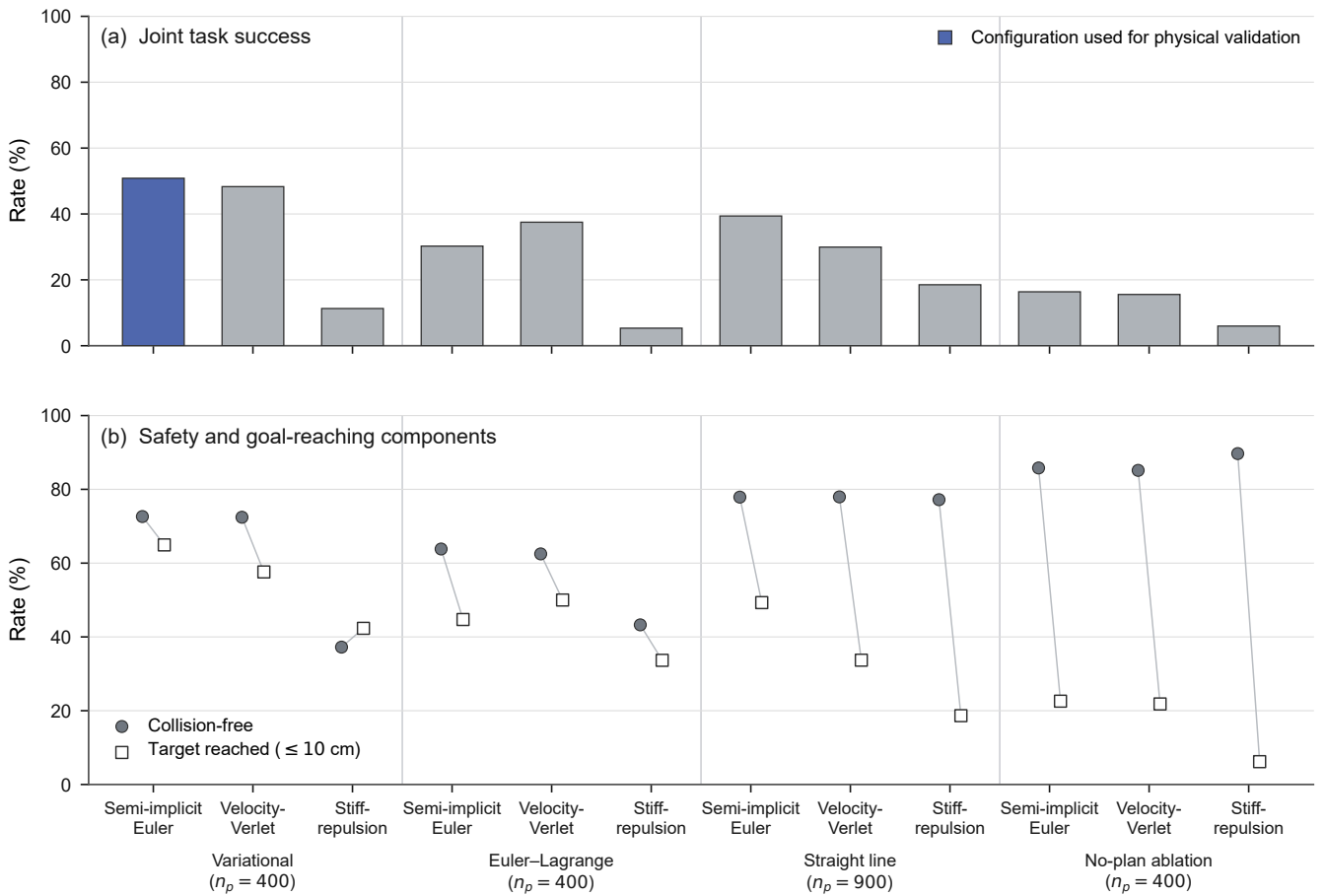


Figure 9. Safety, goal and joint rates over all profiles of the final sampling region, by combination.

idealisations of Section 2.3. They are not evidence about the physical platform, and no combination is recommended here on the strength of them alone.

4.2 Real-time performance of the experimental chain

4.2.1 Sustained loop rate

The closed loop of Figure 2 was required to run at 20 Hz. Over the forty runs of the reported campaign the control loop sustained a median of 19.84 Hz in the single-agent block and 19.86 Hz in the two-agent block, with a standard deviation across runs of 0.05 Hz and 0.09 Hz respectively and a per-run minimum of 19.56 Hz. The rate is thus the most reproducible quantity of

the campaign, at a coefficient of variation of 0.2 % and 0.4 % (Table 12).

Marker detection was complete: the controlled robot was detected in every acquired frame of every run in both blocks. The slowest per-run 99th percentile was 87 ms; isolated loop periods reached 125 ms, and none approached the 0.75 s guard.

4.2.2 Time budget

Table 10 decomposes the mean cost of one control iteration.

Three observations follow, and they are not of equal weight.

First, *the causal correction is computationally negligible*. Computing Equations (18) to (22) takes 0.72 ms, that is 1.4 %

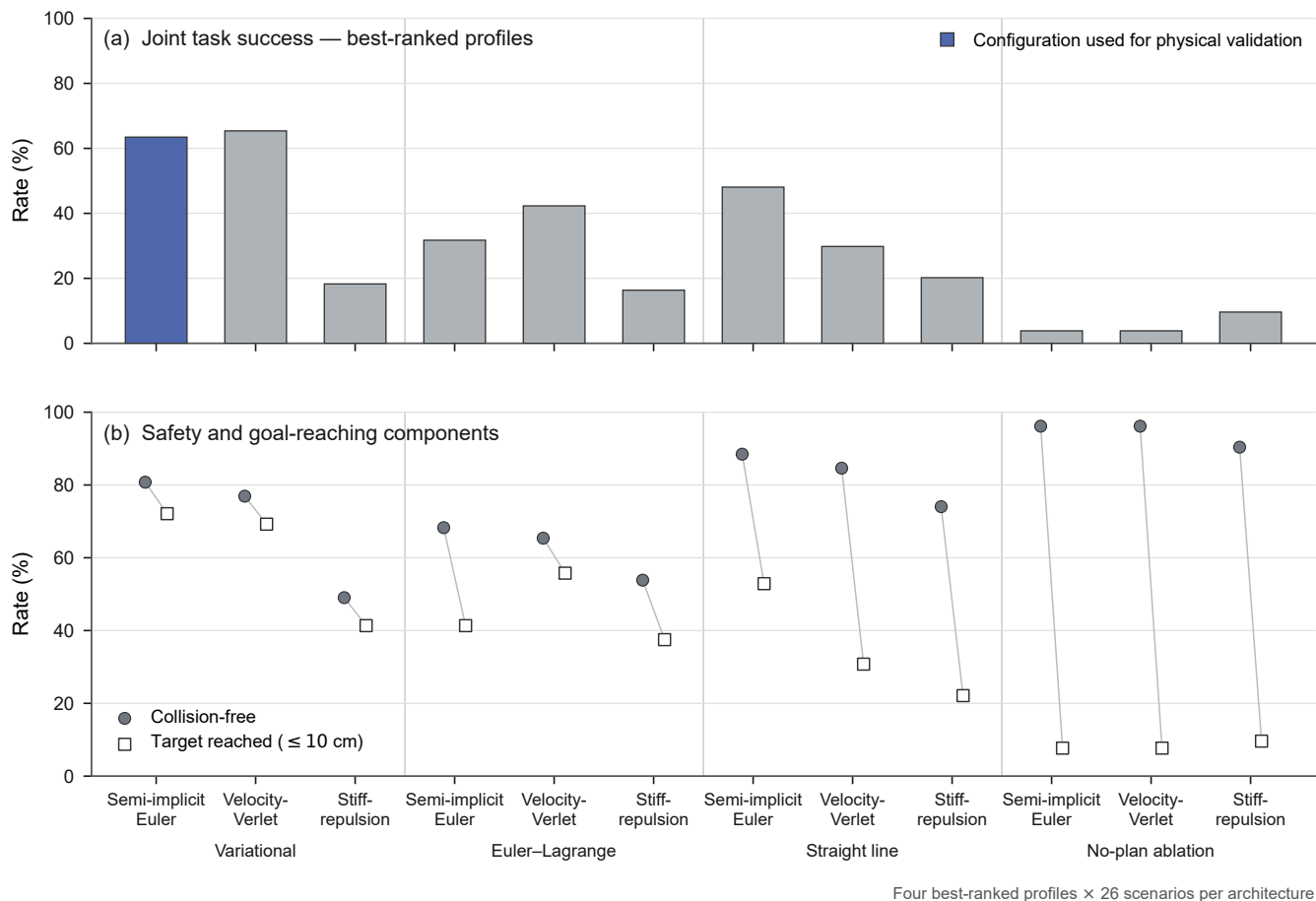


Figure 10. Safety, goal and joint rates over the best-ranked profiles of each combination, evaluated on all twenty-six scenarios.

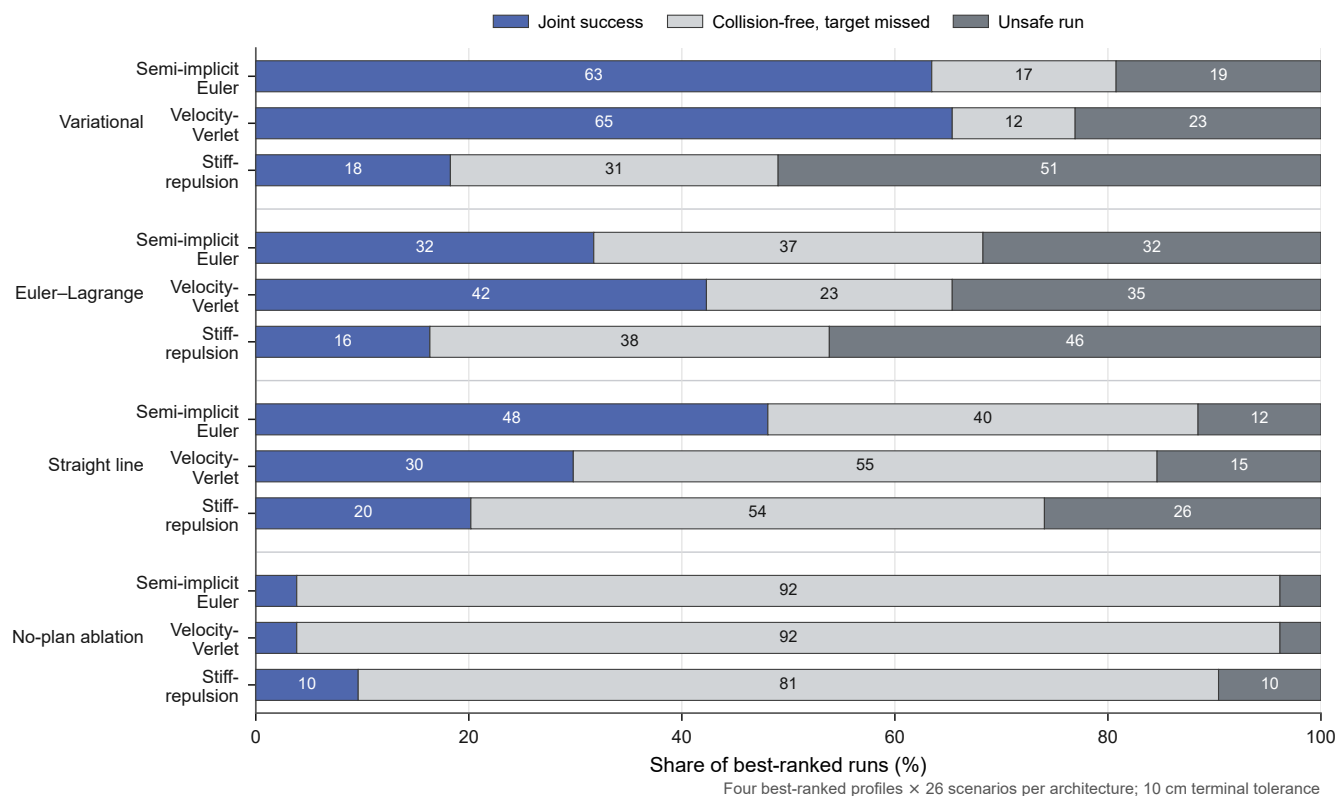


Figure 11. Exclusive decomposition of run outcomes into complete success, safe but short of the goal, and unsafe, by combination.

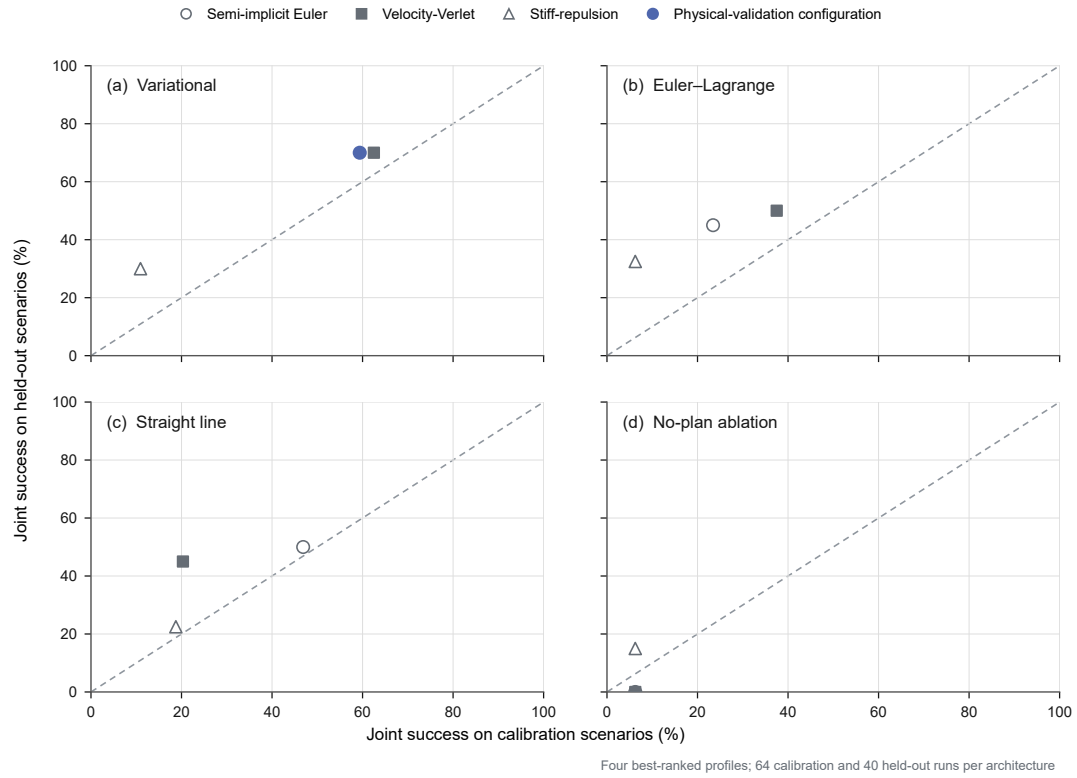


Figure 12. Joint rate on the calibration scenarios against joint rate on the held-out scenarios, for the best-ranked profiles of each combination.

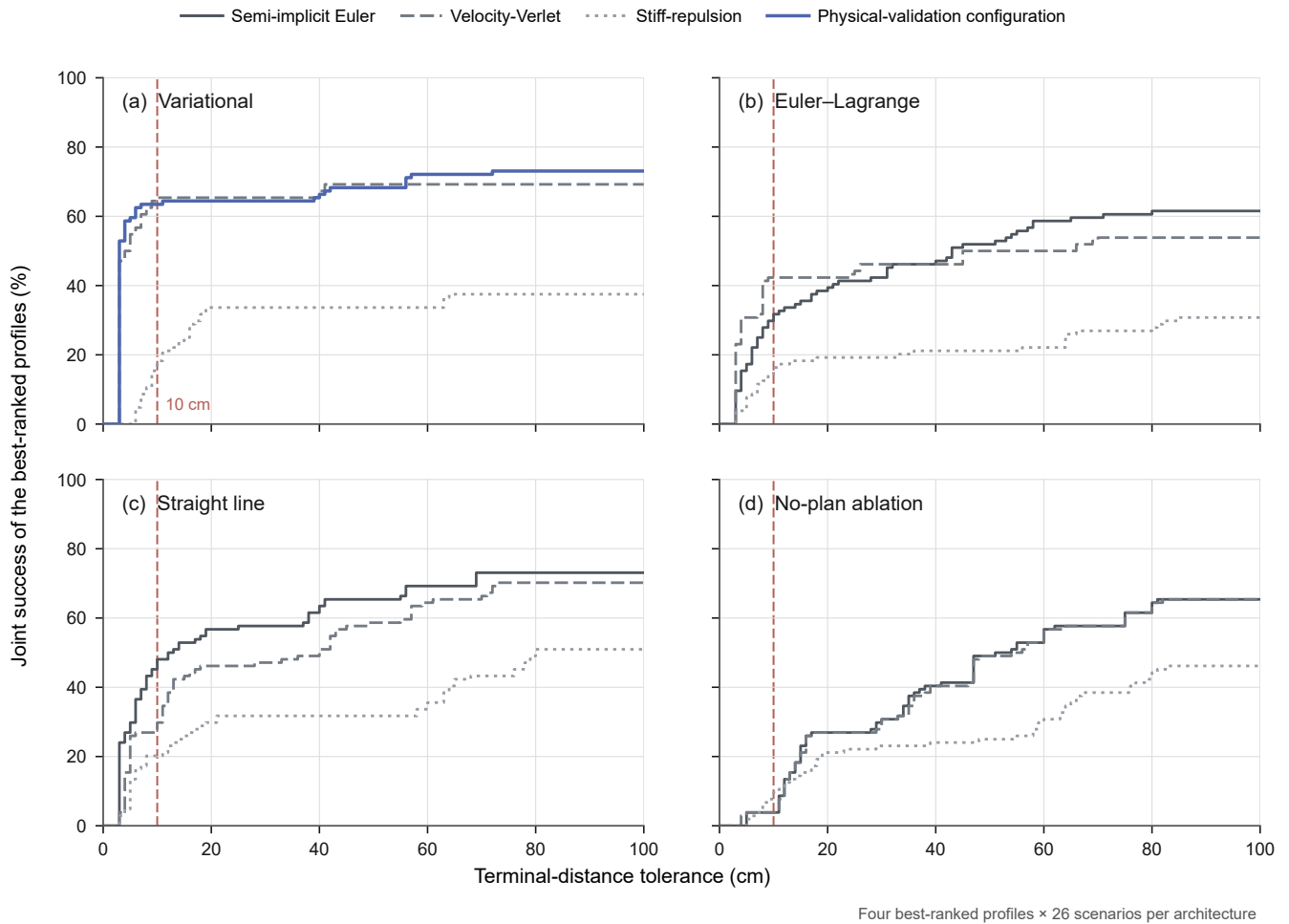


Figure 13. Joint rate of the best-ranked profiles as the terminal tolerance is swept from zero to 100 cm.

Table 10. Real-time budget of one control iteration, averaged over the twenty runs of the single-agent block. Percentages are referred to the 50 ms period of the 20 Hz loop.

Stage	Mean (ms)	Share of budget (%)
Vision, total	32.94	65.9
marker detection	29.07	58.1
image downscaling	2.21	4.4
homography and pose	1.17	2.3
Serial transmission	8.43	16.9
Frame wait	0.83	1.7
Corrector computation	0.72	1.4
Logging	0.20	0.4
Remaining margin	6.88	13.8

of the control period. Whatever limits this system, it is not the decision algorithm. This also means that the cost argument sometimes advanced in favour of simpler correction laws does not apply at this scale: the two-evaluation scheme of Equation (23) would still consume less than 3 % of the budget.

Second, *perception dominates*, at 65.9 % of the period, and within it marker detection alone accounts for 58.1 % of the whole control period. This is where the real-time margin is won or lost.

Third, the serial transmission to the robot costs 8.43 ms, close to 17 % of the budget. This is a cost of physical mechatronic integration that is rarely quantified, and it is of the same order as the entire remaining margin.

Figure 14 completes the table in the one respect a mean cannot address: it shows how each contribution varies from run to run. The dispersion is small and it is concentrated in the vision stage, so the budget of Table 10 describes every run of the block and not merely their average; the residual margin never closes over the twenty runs.

4.2.3 Detection resolution and achieved rate

The 20 Hz figure above is not a property of the hardware alone. An earlier set of runs used the same controller coefficients, identified motion model, requested rate, bench and vision calibration, but performed marker detection on the full acquired frame. The reported campaign instead downscales the frame before detection. Table 11 compares the two sessions; because their scenarios, dates and agent behaviours also differ, the comparison supports an association with processing resolution rather than an isolated causal effect.

Downscaling costs 2.21 ms, while the two sessions differ by 43.74 ms in detection time, a net difference of 41.5 ms per frame. The cost of marker detection is dominated by adaptive thresholding and contour extraction, whose complexity scales with the number of pixels; the pixel count is divided by 2.25 and the measured time by 2.51, which is consistent with a near-linear dependence on image area.

Two consequences deserve emphasis. The achieved rate was 1.75 times higher in the downscaled session. No loss was observed for the controlled-robot marker, which was detected in every frame of both sets; the all-marker detection rate decreased by 0.62 percentage point. The number of frames the loop could not consume falls from about 130 per run to 0.2. This second effect matters beyond throughput, because a loop that cannot keep up with the camera is also a loop that eventually fails to obtain a fresh frame within its real-time deadline.

Two reservations must accompany this comparison. The two sets of runs were not acquired as a designed experiment: they belong to different sessions with different scenarios and different agent behaviours. The timing figures are consistent with the expected effect of the resolution change, but the sessions do not constitute a paired benchmark on the same images; *the difference in outcome between the two sets cannot be attributed to resolution*, and no such attribution is made here. The result is also stated within the conditions of this bench — arena size, camera height and marker size — and does not establish that the same reduction would be harmless at longer range.

4.3 Physical avoidance of moving agents

4.3.1 Exploratory runs and the selection of the reported configuration

Prior to the campaign reported here, twenty-eight attempts produced an archived nominal trajectory, covering twenty-four distinct goal positions. Twenty-six of these attempts have complete execution summaries, covering twenty-two distinct goal positions. The goal was reached in twenty-four of the twenty-six complete runs; the two remaining runs are a progression stall and a safety stop on a momentary marker loss, and neither is a collision. Nine runs recorded an overlap of the conservative bounding discs; no physical contact was observed. No obstacle-configuration count is reported because no archived grouping rule separates intended layouts from small relocation differences. These runs were exploratory and establish feasibility across tested configurations rather than reliability. The configuration reported below was selected among them for its reliability and characterised by repetition. *No figure of this paragraph is used as a result*; the quantitative claims of this section all rest on the repeated campaign described next.

4.3.2 Protocol and perimeter

The reported campaign comprises *forty runs performed on a single day with a single coefficient set* — that of Table 6 — and no retuning between runs. All runs use the variational nominal generator and the semi-implicit Euler corrector, integrating the acceleration $\mathbf{F}/m$ of Equation (22) with m set to the measured mass of the robot. That mass is recorded in the log of every run, together with the acceleration law it enters, so that the dynamic parameter of the corrector is traceable run by run. The moving agents are physical robots driven as described in Section 3.1.4; no virtual agent was injected at any point.

The campaign is organised in two blocks of twenty runs each. Within a block the task is frozen: start, goal and obstacles are held fixed, and the residual dispersion of their measured

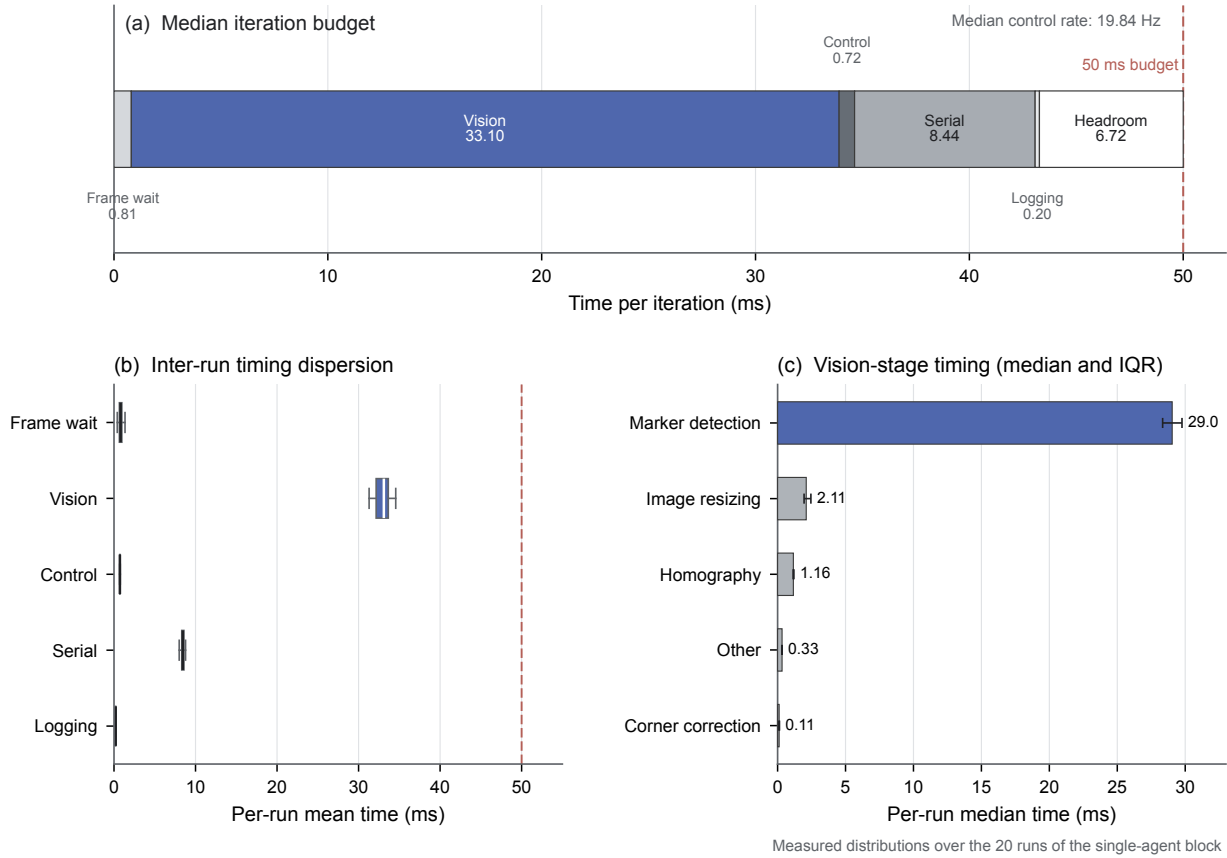


Figure 14. Control-loop time budget over the twenty runs of the single-agent block. (a) Median cost of one iteration by processing stage, against the 50 ms period of the 20 Hz loop. (b) Dispersion of the per-run mean cost of each stage. (c) Median and interquartile range of the vision sub-stages. Panel values are medians across runs and therefore differ by a few hundredths of a millisecond from the means of Table 10.

Table 11. Comparison associated with downscaling before marker detection. Both sets of runs use the same controller coefficients, the same identified motion model, the same requested loop rate and the same vision calibration; they belong to different sessions and also differ in scenario and agent behaviour. A discarded frame is one delivered by the camera and overwritten before the control loop could consume it. The detection rate is averaged over all tracked markers, the controlled robot and the moving agents alike.

Quantity	Full-resolution detection	Downscaled detection
Runs	12	20
Image downscaling (ms)	0.00	2.21
Marker detection (ms)	72.81	29.07
Vision, total (ms)	74.54	32.94
Corrector computation (ms)	0.70	0.72
Achieved loop rate (Hz)	11.35	19.85
Discarded frames per run	129.8	0.2
Marker detection rate (%)	100.00	99.38

positions — at most 1.5 mm in radial deviation for the obstacles — is that of the vision system re-detecting an unchanged scene before each run. The measured start positions lie within 0.7 cm of the block median in every run of both blocks.

What separates the two blocks is stated in full, because the dispersion figures below are compared across them. The second block adds a moving agent, and it also uses a different goal and a different fixed-obstacle layout, so the two blocks are two distinct scenes and not one scene with an extra agent. What is held identical between them is the *machine*: the same corrector coefficients, the same identified motor model, the same reverse-motion thresholds of Section 3.5.2, the same measured mass in the acceleration law, the same requested loop rate, the same vision calibration and the same bounds in the paired simulation.

That alignment is deliberate — the two blocks were re-run together for it — and it is what makes a quantity such as the loop rate or the path-length dispersion comparable from one block to the other, while a quantity tied to the geometry, such as the obstacle clearance, is not.

Applying the criterion of Equation (33), every run of the first block contains exactly one genuinely moving agent and every run of the second contains exactly two, with net agent displacements between 1.02 m and 1.42 m. The perimeter therefore matches the nominal description of each block without exception, which is what makes the dispersion below interpretable.

One run of the second block is excluded from the aggregates and identified here: it was executed before the fixed obstacles were moved to the layout of that block, and therefore carries

the obstacle positions of the first block. Its trajectory is a valid run of a different scene, not a repetition of the task under study, and it is removed on that ground alone — it reached the goal like every other run. The aggregates below accordingly rest on twenty runs of the first block and nineteen of the second.

4.3.3 Outcome and failure modes

The goal was reached in every one of the forty runs, in both blocks, with a terminating condition of `goal_reached` recorded in each run log without exception. No collision, no arena exit, no safety stop on perception loss and no termination by the loop-period guard occurred.

The absence of guard terminations deserves a quantified statement, because the guard it refers to is a configured one. The controller aborts a run whose loop period exceeds 0.75 s. Over the forty runs the slowest per-run 99th percentile of the iteration period is 87 ms. The longest observed loop period was 125 ms, six times below the 750 ms guard. This is an observation on forty runs of a single configuration and not evidence that the mechanism has been eliminated; the point is developed in Section 5.5.

4.3.4 Repeatability

Table 12 reports the dispersion of each measured quantity over the runs that reached the goal, ordered by coefficient of variation.

The ordering is itself the result. *What the machine controls is reproducible; what the environment imposes is not.* The loop rate, the path length and the duration vary by 0.2–5.5 %, whereas the clearance to the moving agents and the tracking error vary by 4.5–28 %. The robot executes the same manoeuvre each time to within a few percent, and the margin it ends up with depends on where the disturbance happens to be.

One entry of that table must not be over-read. The dispersion of the final distance to the goal is small because the run terminates when that distance falls below the tolerance of 0.08 m; it is bounded by the stopping criterion and is not a measure of terminal accuracy. The obstacle clearance must not be compared across the two blocks either: the fixed obstacles occupy different positions in the two scenes, so the two columns of that row describe different geometries.

The two-agent block exhibited a duration 2.72 s longer and a maximum tracking error 6.4 times larger than the one-agent block; its coefficients of variation for duration and path length were four to five times larger. Its loop rate remained essentially unchanged and its fraction of corrected commands was slightly lower. Because the blocks also differ in target, obstacles and agent command, these contrasts are descriptive rather than an identified effect of adding a second agent.

Figures 15 and 16 give the geometric counterpart of that table, one figure per block. They superimpose *all* the runs entering the aggregates rather than a selected one, which is what makes the dispersion of Table 12 legible: the executed paths form a narrow bundle around a single manoeuvre, and the clearance histories share one shape, with the run-to-run spread concentrated in the depth of the closest approach and in the recovery that follows it. The tightening of the bundle after the closest approach is the same ordering as in the table — the trajectory is reproduced, the margin is not.

4.3.5 Clearances

Clearances require a careful statement, because they are the quantity most easily misread. As defined in Section 3.7.2,

they are measured between conservative bounding discs of radius 0.130 m, and a negative value indicates that those discs overlapped, not that the robots touched. *No physical contact occurred at any point during the campaign.*

No negative clearance was recorded in any run of the campaign. Both the agent and the obstacle clearance remained strictly positive throughout all forty runs: the minimum agent clearance observed was 10.6 cm in the single-agent block and 6.3 cm in the two-agent block, and the minimum obstacle clearance 29.7 cm and 49.6 cm respectively.

This is a markedly better safety record than the exploratory runs of Section 4.3.1, and the comparison should be read with care: those runs spanned twenty-four different geometries, several of them deliberately more constrained than the one repeated here. *The claim supported by this campaign is that the reported configuration is executed safely and repeatedly, not that the method maintains a positive clearance in general.* A soft-force corrector provides no such guarantee, which is a structural property of potential-field methods with bounded acceleration rather than an artefact of tuning; the point is developed in Section 5.5.

4.3.6 Comparison with the preceding campaign

An earlier campaign of the same two blocks was run on the preceding day with an identical start, an identical goal and the same coefficient set, but without the robot mass entering the corrector. Comparing the two is informative, and the comparison must be qualified block by block, because the two blocks did not change by the same amount.

In the two-agent block, the mass is the only control parameter that differs between the two campaigns: the gains, the perception radius, the look-ahead, the motor parameters, the reverse-motion thresholds and the simulation bounds are identical. Three differences of a different nature remain — the fixed obstacles were physically displaced by 4.8 cm and 10.3 cm, the two agent commands were exchanged between the two physical agents, and the run excluded above. In the single-agent block the comparison is weaker, and it is weaker by design: that block is the one that was brought into line with the other. Its reverse-motion thresholds moved from 110° and 100° to the values of Section 3.5.2, already in use in the two-agent block, and the acceleration bound of its paired simulation moved from 0.8 m/s² to the 1.0 m/s² already in use there. Four control parameters therefore changed at once in that block. The change was deliberate — it is what places the two blocks on a common motor and simulation configuration, as declared in Section 4.3.2 — and its price is that the single-agent block cannot be compared with its predecessor parameter by parameter.

No causal attribution to the mass is therefore made here. What the two campaigns support is a statement about the protocol rather than about the parameter: every run of the new campaign reached its goal, against thirty-seven of forty previously, and the dispersion of the quantities the controller governs is equal or lower — the coefficient of variation of the loop rate falls from 1.0 % to 0.2 % and that of the path length from 1.9 % to 0.9 % in the single-agent block, on runs that reached the goal in both cases. The measured mass is reported here as a documented property of the configuration, traceable in every run log, and not as an explanatory variable.

Table 12. Dispersion over repeated runs of a frozen task. Values are medians with the coefficient of variation in brackets, ordered from the most to the least reproducible quantity of the single-agent block.

Quantity	One moving agent ($n = 20$)		Two moving agents ($n = 19$)	
	Median	CV	Median	CV
Loop rate (Hz)	19.839	0.2 %	19.863	0.4 %
Path length (m)	1.931	0.9 %	2.336	3.6 %
Duration (s)	7.721	1.1 %	10.439	5.5 %
Obstacle clearance (m)	0.310	1.9 %	0.536	3.7 %
Final distance (m)	0.073	4.5 %	0.072	7.4 %
Agent clearance (m)	0.117	4.5 %	0.094	13.0 %
Corrected commands (%)	85.2	7.3 %	79.9	3.6 %
Maximum tracking error (m)	0.055	28.2 %	0.351	12.1 %

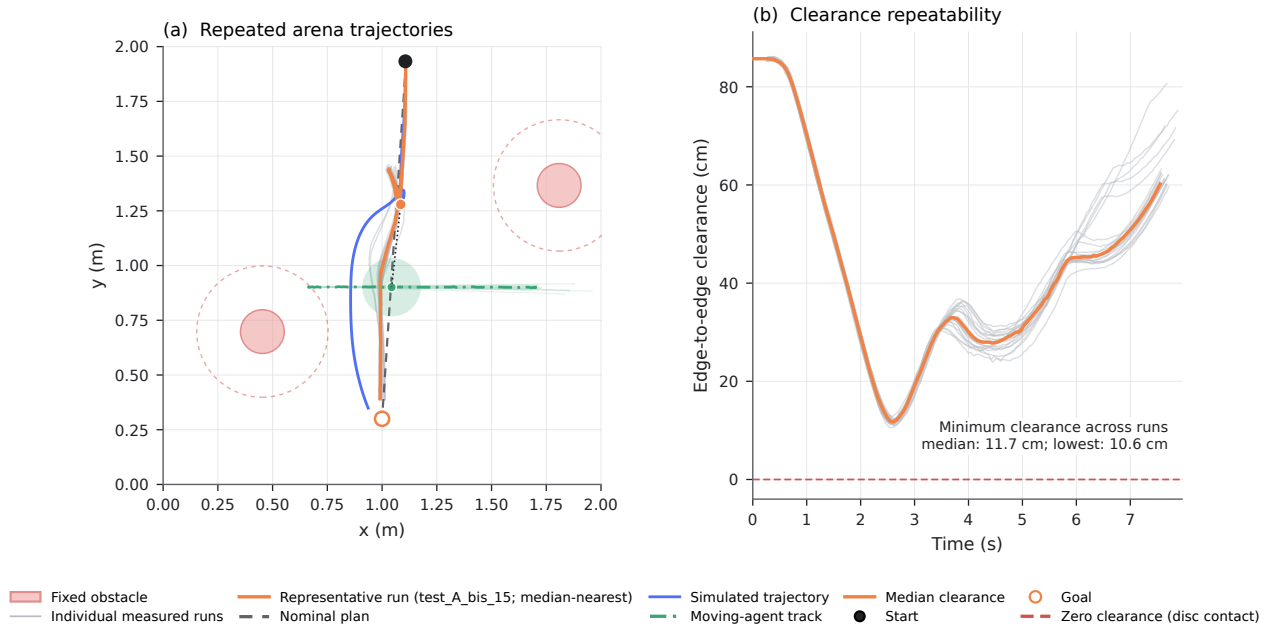


Figure 15. Physical avoidance of one moving agent, over the twenty runs of the block. (a) Arena view: the twenty measured paths, the run closest to the block medians, its paired simulation, the nominal plan, the moving-agent track and the closest approach of that run. (b) Edge-to-edge clearance to the agent over time for every run, with the median. Clearances are measured between conservative bounding discs, so the zero line marks contact between those discs and not between the robots.

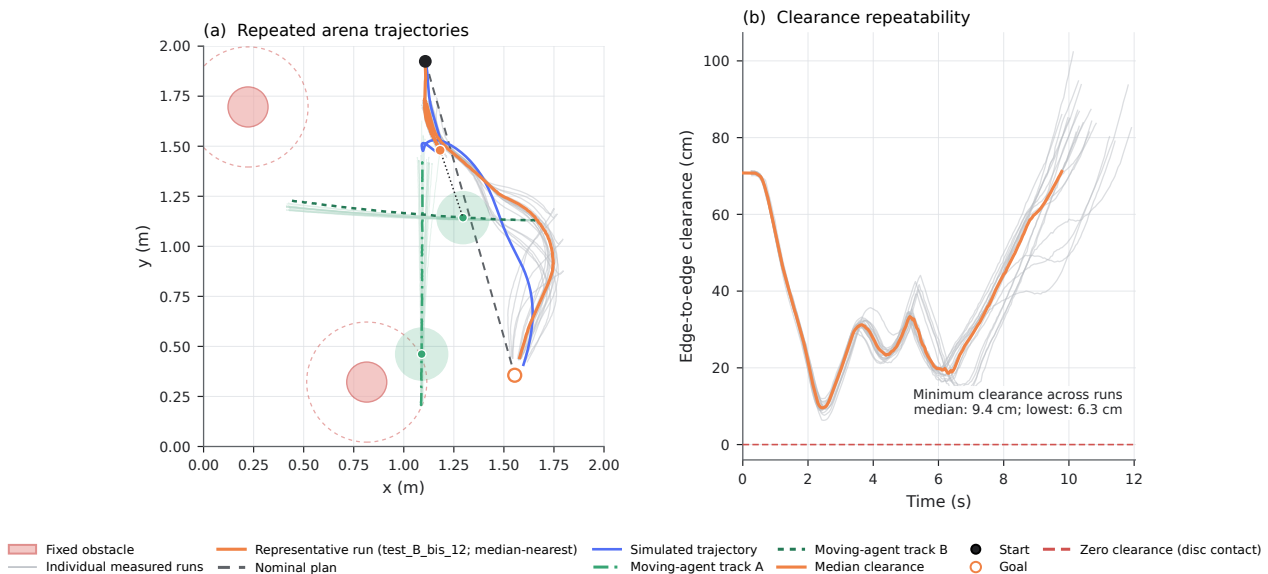


Figure 16. Physical avoidance of two simultaneously moving agents, same layout as Figure 15, over the nineteen runs retained in the block aggregates. The clearance reported in (b) is the smallest of the two agent clearances at each instant.

4.4 Simulation-versus-experiment agreement

4.4.1 Path-level agreement

Every run was paired with a simulation of the same scenario and compared using the measures of Equation (34). Table 13 reports the outcome for the runs that reached the goal.

The median mean departure between the simulated and the measured path is 5.8 cm with one moving agent and 5.3 cm with two, and the executed path is a median 1.090 and 1.169 times longer than the simulated one. The two blocks agree closely on the mean departure and differ mainly in the maximum, which is consistent with a larger single manoeuvre in the two-agent block rather than a generally worse fit.

The nominal trajectory planned by the real-time runner and the one recomputed inside the simulation are generated from the same archived scene and nominal cost. Their geometric difference is not quantified here because the original pairing convention was not archived. The path-level departures below therefore combine any residual planning difference with execution effects.

In the BIS campaign, the simulation and the physical runner share the same kinematic bounds: 0.25 m/s in speed and 1.0 m/s² in acceleration. The archived manifest field names retain a `simulation_` prefix, but the campaign applies these globally to both sides of the comparison.

4.4.2 Robustness of the length measurement

Because the measured path is reconstructed from localisation samples at 20 Hz, its length is inflated by localisation noise: a noisy polyline is longer than the smooth curve it samples. Retaining one sample in ten of the measured polyline reduces its median length by 3.7 % and 3.2 % in the two blocks, and brings the length ratio from 1.090 to 1.044 and from 1.169 to 1.128.

Part of the measured elongation is therefore a sampling artefact, and part of it is not. About four percent of excess length survives decimation in the single-agent block and about thirteen percent in the two-agent block, so the elongation cannot be dismissed as a measurement effect, and neither can the raw ratios be read as pure execution error. The decimation factor is itself arbitrary, and the figures above quantify the sensitivity of the length measurement rather than resolve it.

4.4.3 Status of the digital twin

Within the configuration studied, the simulation reproduces the executed path to within a mean departure of about 5 cm and reproduces the terminal condition of every run. It does not reproduce the path length: the measured path remains longer than the simulated one by a margin that survives the sampling correction, and that margin is larger with two moving agents than with one.

Two qualifications are essential. First, this agreement is established on *one* task geometry per block, repeated; it says nothing about geometries on which the twin has not been tested, and the exploratory runs of Section 4.3.1 are not paired here for that reason. Second, the twin remains a holonomic point-mass model: the agreement reported above is an empirical result on this configuration and not evidence that the model captures the non-holonomic constraint. The candidate causes of the remaining mismatch are examined in Section 5.3.

5 Discussion

Section 4 presents two distinct sets of results, which are interpreted separately in this discussion. The first comes from

simulation and compares twelve solver-controller combinations. Its purpose is to compare the behaviour of these combinations under the same numerical framework and to motivate the choice of one architecture for further study. The second comes from the physical experiments, where only that selected architecture is tested repeatedly on the real robotic platform. These experiments therefore characterise the behaviour, repeatability, and practical limitations of one specific implementation on hardware.

The two studies should not be interpreted as a direct simulation-to-experiment validation. The physical campaign does not test the eleven other solver-controller combinations considered in simulation, so it cannot confirm the complete ranking or comparison obtained numerically. Conversely, the simulation study does not reproduce all the effects present on the real system and is therefore not sufficient to validate the experimental behaviour. The two parts are instead complementary: simulation provides a broad comparison that supports the architectural choice, whereas the experiments provide a detailed characterisation of the selected architecture under real operating conditions.

Despite this separation, both studies rely on the same methodological question: what should be counted as a successful run? In both simulation and experiment, safety alone is not sufficient, and reaching the goal alone is not sufficient either. The discussion therefore starts from the joint-success criterion used throughout the paper, before interpreting the simulation and experimental results separately.

5.1 Interpretation of the joint-success criterion

The joint indicator of Equation (4) was introduced in Section 2.2 on the grounds that safety and arrival are individually degenerate objectives. The results support that choice in a stronger form than anticipated, because both degenerate behaviours were actually observed rather than merely predicted.

The failure modes separate into two families. A *progression failure* leaves the robot safe but short of the goal: the attraction and repulsion terms of Equation (18) balance before arrival and the robot stops in a locally safe configuration. A *safety failure* is the opposite: a repulsion law stiff enough to react strongly near contact keeps the corrector saturated and degrades terminal progress, so that improving the reaction to proximity costs the ability to arrive.

The practical consequence is that a high safety rate, reported alone, is not evidence of a good controller — it is compatible with a controller that never moves. Selecting an architecture on safety alone would systematically favour the least useful one. This is the reason the joint criterion is used throughout this work, and it is a reporting recommendation as much as a design one. Figure 11 is the form in which we would recommend reporting such a comparison: by decomposing each combination into complete success, safe but short of the goal, and unsafe, it makes the two failure families visible side by side, which a pair of aggregate rates does not.

One run of the exploratory set carries this argument from simulation to hardware. A run that reached its maximum duration ended with a clearance of 81.2 cm and a departure from the nominal path of 2.4 cm: it was entirely safe, it was close to its plan, and it did not arrive. Under a safety-only criterion this run would be counted as a success.

Table 13. Geometric agreement between simulated and measured paths over the repeated campaign. All values are medians across runs.

Configuration	Runs	Mean departure (cm)	Maximum departure (cm)	Hausdorff (cm)	Length ratio
One moving agent	20	5.81	14.26	14.26	1.090
Two moving agents	19	5.31	18.00	18.00	1.169

5.2 Value of the global nominal plan

The architecture of this work separates an offline global plan from a causal local correction, and Section 4.1 makes it possible to ask what each layer contributes. The answer is not the one the separation suggests.

The decisive comparison is not between the two nominal generators but between having a plan and having none. The ablation that removes the global plan entirely, keeping the same corrector and the same tuning protocol, falls to a joint rate of 3.85–9.62 % over its best-ranked profiles, against 20.19–48.08 % for the same three correctors driven by a straight line drawn through the obstacles. *A plan that ignores the obstacles is worth far more than no plan at all.* This is consistent with the failure mode identified in Section 4.1.2: what the plan supplies is not a collision-free path — the straight line is not one — but a spatially extended attraction that keeps the corrector progressing instead of settling where the local forces balance.

Against that gap, the further gain from a plan that does account for the obstacles is comparatively modest: 65.38 % and 63.46 % for the two variational combinations, some fifteen to seventeen points above the straight-line baseline at the 10 cm reference tolerance. Two qualifications make the difference harder rather than easier to dismiss. The baseline was granted more than twice the exploration budget — 900 profiles against 400 — so a tuning artefact would work in its favour and the reported gap is the conservative side of the estimate. And the two variational combinations are also the two that transfer best to the held-out scenarios, both reaching 70 % against 50 % for the baseline, so the advantage is not an artefact of the scenarios on which the profiles were selected.

The interpretation we retain is therefore a division of roles rather than an attempt to assign a precise fraction of the performance to each component. The present results do not allow us to quantify how much of the overall performance comes from the global plan and how much comes from the local corrector. Instead, they suggest that the two components serve different and complementary functions.

The global plan provides a spatially extended direction of progression towards the goal, whereas the local corrector reacts to the current positions of nearby obstacles and agents. The global plan does not directly perform the real-time avoidance. Rather, it reduces the burden placed on the local corrector: without such a plan, the corrector must both avoid nearby obstacles and, using only local attractive and repulsive forces, implicitly determine how to continue towards the goal. This can lead to configurations in which these local forces balance and the robot remains safe but stops progressing. By providing a global direction of progression, the nominal plan reduces the need for the corrector to discover such a global detour from local information alone.

This interpretation is consistent with the success rates reported in Table 9, but it should not be read as a quantitative measurement

of the individual contribution of each layer. This division of roles follows directly from the architecture: the nominal planner has no access to the future motion of the other agents, whereas the causal corrector is the component that observes their current states and modifies the trajectory in response. Without this correction, the rover would simply follow its nominal trajectory without reacting to the moving agents. The ablation results further show that the converse is also problematic: when the corrector is deprived of a global nominal plan, local avoidance alone frequently fails to maintain progress towards the goal.

Two boundaries apply to this subsection and are not rhetorical. The first is that the ordering above is stated at one terminal tolerance, and Figure 13 shows that it does not survive a change of that requirement: at 100 cm the straight-line baseline matches the best variational combination. That is not a detail, it sharpens the reading just given. What the variational plan buys is not a better ability to get past the obstacles — the baseline gets past them too — but a better ability to finish, and the fifteen-to-seventeen-point gap measured at 10 cm is the price of terminal precision rather than the price of avoidance. And these are simulation results obtained under the idealisations of Section 2.3: they motivate the architecture that was then taken to hardware, and they are not evidence about it. The physical campaign of Sections 4.2 to 4.4 uses a single combination throughout and measures nothing about the eleven others.

5.3 Sources of discrepancy between plan, twin and reality

Section 4.4 established that the executed and the simulated paths agree to a mean departure of about 5 cm, and that the executed path is nevertheless the longer of the two by a margin that survives the sampling correction. The twin therefore reproduces *where* the robot goes and not *how much ground it covers getting there*, and the second half of that statement is what the mechanisms below have to account for. The agreement holds on the repeated configuration and should not be extrapolated; the mechanisms are ranked by how well the present data supports them.

The dominant structural difference is that *the simulation integrates holonomic point dynamics while the robot is non-holonomic*. The corrector of Section 3.4 produces an acceleration in an arbitrary direction, and the simulation applies it directly. The physical robot cannot: it must reorient before translating, it alternates between turning in place and driving along an arc, and its commands are smoothed before transmission. None of that is present in the simulated model. A rapid heading change, which costs the simulation nothing, costs the robot both time and path length. The signature is visible in Figures 15 and 16: the simulated and the measured paths leave the start together and arrive together, and separate over the segment where the avoidance occurs. It is also consistent with the two aggregate measures of Table 13, since the two-agent block leaves the mean departure essentially unchanged while raising the maximum departure and the length ratio — the

behaviour of a discrepancy produced by manoeuvres rather than distributed along the path.

Three measured properties of the platform compound this effect. The command-to-motion response is affine rather than linear through the origin, and the corrector inverts a fixed linear model; the mismatch is largest exactly where small corrections are requested. The left–right response is asymmetric, and this asymmetry is not compensated by the controller, which applies a symmetric model. And the drivetrain has no wheel encoders, so nothing detects or corrects a wheel-level deviation between two camera frames.

The real-time runner and the simulation share the same archived scene and nominal cost, but their nominal trajectories are recomputed independently and their difference is not quantified here because the pairing convention was not archived. A residual planning contribution to the measured path departure therefore cannot be excluded.

Perception latency contributes a share that the budget of Table 10 makes it possible to bound from below, and the bound is larger than one might expect. A frame is on average 17 ms old when the control loop takes it, and reaches 35–50 ms at the per-run worst case; the command derived from it is emitted a further 42 ms later, once detection, pose estimation, the correction itself and serial transmission have been paid. The corrector therefore acts on a state at least 60–90 ms old, during which the robot has travelled between one and three centimetres at the speeds observed here.

That figure is a floor in two respects, and both are declared rather than estimated. The measured age is counted from the instant the acquisition thread completed the reception of a frame, so exposure, sensor readout and network transport to the host are outside it; and the loop contains steps between the instrumented ones that are not timed. Set against a mean departure of about 5.5 cm, a lower bound of one to three centimetres is a contribution of the same order as the effect to be explained rather than a negligible one. It does not account for the discrepancy on its own, and it cannot be dismissed from it either — which is a reason to instrument the sensor-to-command path end to end before attributing the residual to the non-holonomic constraint alone.

One potential source can be *excluded* on the evidence available: the moving-agent trajectories used in simulation are the measured tracks rather than idealised straight lines, so the lateral drift of the open-loop agents is already accounted for and cannot be invoked here.

The BIS campaign applies the same kinematic bounds on both sides of the comparison: 0.25 m/s and 1.0 m/s². Kinematic-bound mismatch therefore does not explain the observed elongation.

A twin incorporating the non-holonomic constraint, the measured affine command-to-motion response, and the actuation asymmetries is therefore a natural next step. This extension is realistic, since the main parameters required to refine the model have been measured in this work. A complementary direction would be to use a robotic platform providing greater access to its low-level actuation and control layers, including wheel encoders, motor commands, and the embedded control loop. Such a platform would make it possible to identify the command-to-motion dynamics more accurately, implement direct wheel-level feedback, and better separate discrepancies caused by the proposed high-level corrector from those introduced by the underlying hardware and low-level controller.

5.4 What the real-time budget shows

The time budget of Table 10 contains a result that is worth separating from the rest, because it inverts a common assumption. The causal correction — the part of the system that carries the scientific contribution — consumes 1.4 % of the control period. Marker detection consumes 58 %. *In this system the algorithm is free and the perception is expensive.* Figure 14 shows that this is a property of the chain rather than of a favourable run: the ranking of the stages is identical in every run of the block, and the stage that dominates is also the one that disperses.

This has a direct design consequence, quantified in Section 4.2.3: the sessions using downsampled detection were associated with an achieved loop rate 1.75 times higher. No loss was observed for the controlled-robot marker, while all-marker detection decreased by 0.62 percentage point. A paired offline benchmark on the same images is needed to isolate the effect of resolution.

The second consequence concerns failure. A loop that cannot keep pace with the camera accumulates discarded frames — about 130 per run in the full-resolution sessions against 0.2 in the downsampled sessions — and eventually misses its real-time deadline. Throughput and timing robustness are therefore linked objectives in this chain.

The boundary of that argument must be stated, because it is the one a reader is most likely to cross unaided. The comparison reports a strong association between downscaling and lower processing time, but it is not a paired benchmark on the same images. It is not shown to buy *avoidance performance*. The two sets of runs compared in Section 4.2.3 differ in scenario and in agent behaviour as well as in resolution, and the experiment that would separate the two — the reported configuration re-run at the lower rate, everything else held fixed — was not performed. *No link between the perception rate and the quality of the avoidance is claimed anywhere in this paper*, and the reader should not supply one.

A more general point follows for vision-in-the-loop mechatronic systems. The serial transmission to the robot costs 8.43 ms, close to 17 % of the period and more than the entire remaining margin. Integration costs of this kind are seldom reported, yet in this system they are an order of magnitude larger than the control computation. They also carry the tail: the only two runs the preceding campaign lost to the loop-period guard were lost inside that same serial write, as reported in Section 5.5.

The link between the Raspberry Pi and the robot is at once the second-largest cost of the budget and the least predictable part of it, which is an argument for reporting integration timings and not only algorithmic ones.

5.5 Limitations

The limitations below are ordered by how much they constrain the conclusions.

No formal guarantee of non-collision. The corrector uses soft repulsive forces with bounded acceleration. Nothing in its formulation enforces a minimum separation. The reported campaign happens to record no negative clearance at all (Section 4.3.5), but that outcome is a property of the forty runs performed, not of the formulation: the exploratory runs of Section 4.3.1 produced a negative clearance in nine of their twenty-six runs. In this paper, each of these events is classified as a collision under the geometric safety criterion, because the conservative bounding discs overlapped. Physical contacts were also observed during the exploratory tests. The two statements are not interchangeable: because the discs conservatively bound the chassis, a negative clearance does not necessarily imply physical contact. This distinction must not obscure the central limitation: *this architecture does not guarantee a minimum clearance*, and no experimental campaign can supply a guarantee that the formulation does not contain. A clean record over a repeated configuration is evidence about that configuration and nothing more.

Local minima and progression stalls. The same soft-force structure admits configurations in which attraction and repulsion balance before the goal. This failure mode is visible in simulation, where it dominates the ablation without a global plan, and it occurred once on hardware, in the exploratory runs. It did not occur in the reported campaign, which is a statement about the geometry that was repeated and not about the method: it is not a tuning defect but a known property of potential-field formulations, and it is the principal argument for combining this architecture with a global replanning or graph-search stage.

Single bench, single campaign, one or two agents. All the results reported here come from one arena, one hardware set and a single coefficient set, and the forty runs of the campaign were acquired on a single day. Avoidance was exercised against *one or two* moving agents; larger groups were examined in simulation only, and no claim of physical multi-robot validation beyond two agents is made. The reported rates are frequencies over a finite set of runs, not probabilities, and carry no confidence interval.

Repeatability was measured on one configuration. The campaign of Section 4.3.4 repeats a frozen task forty times and quantifies its dispersion, but it does so on a single geometry per block. It supports a claim of repeatability and *does not* support a claim of generalisation across configurations. The exploratory attempts with archived nominal trajectories cover twenty-four distinct goal positions, but only twenty-two among the runs with complete execution summaries. They do not provide a repeated-geometry design, so neither set establishes both properties. Extending the repeated protocol to several

geometries is the natural next step, and it requires only bench time.

The final distance is bounded by the stopping condition. Successful runs terminate when the distance to the goal falls below 0.08 m, so the recorded final distances indicate where that condition was triggered. They are not a measure of positioning accuracy and are not used as one.

The corrector only reacts. It reads the current state of the observed agents and has no model of where they are heading. Its reaction therefore begins only once an agent is already close, and a fast agent on a converging course leaves it correspondingly less room. Adding a short-horizon predictive term is the most direct route to improvement, and it can be done without breaking causality as defined in Section 2.3, since a prediction built from past observations reads no future state.

The twin is validated on one configuration only. The agreement reported in Section 4.4 is established on the repeated task and on no other. The twin remains a holonomic point-mass model, and nothing here shows that the agreement survives on a geometry demanding sharper heading changes. The BIS simulation and physical runner use the same speed and acceleration bounds, but the residual disagreement in path length cannot be attributed to a single mechanism.

Modelling assumptions. Motion is planar; the robot and the physically diverse obstacles are represented by discs in the model; and the nominal kinematic limits are soft penalties rather than hard constraints, so the nominal trajectory carries no formal guarantee that they are respected.

Sensing. The overhead camera is the only feedback in the entire chain, and its loss is the loss of the control loop. No run of the reported campaign lost the controlled marker, but one exploratory run was terminated by the safety layer after a momentary marker loss, so the mode is real and merely absent here. Residual localisation error is 11.6 mm on average and 23.1 mm at worst. Against the smallest agent clearance of the reported campaign, 6.3 cm, the worst case is roughly a third of the margin: it is not negligible, and every clearance reported in this paper should be read as carrying an uncertainty of that order. The exploratory runs came far closer to the obstacles than that, which is where a localisation error of this size would cease to be a correction and become the measurement.

6 Conclusion and perspectives

This study assessed an architecture separating a variational nominal trajectory computed offline on the static scene from a causal local corrector driven by the present state of moving agents. No exact future agent state is queried; nominal look-ahead therefore remains causal. In simulation, twelve combinations, independently calibrated in their parameter spaces, differed in joint success. Safety alone was misleading: the no-plan ablation reached 96.15 % safety but only 3.85 % joint success with two correctors because of safe stalls far from the goal. This supports the core of H1 within this protocol, not a universal ranking: selection used one seed, unequal final-region budgets (900 straight-line profiles versus 400 otherwise), and

quarantine over six to fourteen scenarios. Four retained profiles per combination were compared on the common twenty-six scenarios; these frequencies are not probabilities and have no confidence intervals.

H2 is supported as a progression result. At the 10 cm tolerance, the leading variational combinations reached 65.38 % and 63.46 % joint success, compared with 48.08 % for the best straight-line combination. At 100 cm, however, the straight line with the semi-implicit Euler corrector reached 73.08 %, equal to the best variational combination. Thus, in the scenarios studied, the variational nominal trajectory improved progression and terminal precision rather than avoidance in general; the protocol cannot apportion performance quantitatively between the two layers.

H3 is supported within its tested domain by the physical campaign of the variational generator and semi-implicit Euler corrector, with coefficients unrelated to the comparison. All forty same-day runs with one coefficient set reached the goal, twenty per frozen task; the tasks also differed in goal, obstacles, and agent instructions. They shared the hardware–software chain, corrector parameters, mass, reverse thresholds, visual calibration, and kinematic bounds. A second-block run excluded from aggregates for retaining the preceding obstacle layout also reached the goal. No physical contact, negative clearance, arena exit, or safety stop was recorded, and the smallest moving-agent margins were 10.6 cm and 6.3 cm. Median loop rates were 19.84 Hz and 19.86 Hz; correction used 0.72 ms (1.4 % of the period), whereas perception dominated. These observations are confined to this domain and give no formal separation guarantee. Final distance reflects the 0.08 m stopping criterion. With identical 0.25 m/s and 1.0 m/s² bounds, the paired simulation reproduced path geometry approximately (mean departures 5.81 cm and 5.31 cm) but not length: ratios of 1.090 and 1.169 became 1.044 and 1.128 after tenfold decimation. The residual may combine non-holonomic motion, affine asymmetric actuation without encoders, command smoothing, switching, latency, and unquantified differences between recomputed nominal trajectories; no single cause is identified.

Future work should enrich the twin with non-holonomic motion, the measured affine response, actuation asymmetry, command smoothing, and fuller latency models; instrument the end-to-end sensor–command chain; and add wheel encoders or use a low-level-access platform. Causal short-horizon anticipation from past observations and graph-search initialisation or global replanning could address late reactions and local minima. The protocol should then be repeated across geometries, days, seeds, and platforms before progressively extending physical validation to larger groups; configurations beyond two moving agents have so far been studied only in simulation.

References

- [1] Roland Siegwart, Illah R. Nourbakhsh, and Davide Scaramuzza. *Introduction to Autonomous Mobile Robots*. MIT Press, Cambridge, MA, 2nd edition, 2011.
- [2] Sean Wilson and Magnus Egerstedt. The robotarium: A remotely-accessible, multi-robot testbed for control research and education. *IEEE Open Journal of Control Systems*, 2:12–23, 2023.
- [3] John T. Betts. Survey of numerical methods for trajectory optimization. *Journal of Guidance, Control, and Dynamics*, 21(2):193–207, 1998.
- [4] Nathan Ratliff, Matthew Zucker, J. Andrew Bagnell, and Siddhartha Srinivasa. CHOMP: Gradient optimization techniques for efficient motion planning. In *Proceedings of the 2009 IEEE International Conference on Robotics and Automation (ICRA)*, pages 489–494, 2009.
- [5] Mrinal Kalakrishnan, Sachin Chitta, Evangelos A. Theodorou, Peter Pastor, and Stefan Schaal. STOMP: Stochastic trajectory optimization for motion planning. In *Proceedings of the 2011 IEEE International Conference on Robotics and Automation (ICRA)*, pages 4569–4574, 2011.
- [6] John Schulman, Yan Duan, Jonathan Ho, Alex Lee, Ibrahim Awwal, Henry Bradlow, Jia Pan, Sachin Patil, Ken Goldberg, and Pieter Abbeel. Motion planning with sequential convex optimization and convex collision checking. *The International Journal of Robotics Research*, 33(9):1251–1270, 2014.
- [7] Chady Ghnatios, Daniele Di Lorenzo, Victor Champaney, Amine Ammar, Elias Cueto, and Francisco Chinesta. Optimal trajectory planning combining model-based and data-driven hybrid approaches. *Advanced Modeling and Simulation in Engineering Sciences*, 11:10, 2024.
- [8] Oussama Khatib. Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, 5(1):90–98, 1986.
- [9] Yoram Koren and Johann Borenstein. Potential field methods and their inherent limitations for mobile robot navigation. In *Proceedings of the 1991 IEEE International Conference on Robotics and Automation (ICRA)*, volume 2, pages 1398–1404, 1991.
- [10] Elon Rimon and Daniel E. Koditschek. Exact robot navigation using artificial potential functions. *IEEE Transactions on Robotics and Automation*, 8(5):501–518, 1992.
- [11] Johann Borenstein and Yoram Koren. The vector field histogram — fast obstacle avoidance for mobile robots. *IEEE Transactions on Robotics and Automation*, 7(3):278–288, 1991.
- [12] Dieter Fox, Wolfram Burgard, and Sebastian Thrun. The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine*, 4(1):23–33, 1997.
- [13] Paolo Fiorini and Zvi Shiller. Motion planning in dynamic environments using velocity obstacles. *The International Journal of Robotics Research*, 17(7):760–772, 1998.
- [14] Jur van den Berg, Ming C. Lin, and Dinesh Manocha. Reciprocal velocity obstacles for real-time multi-agent navigation. In *Proceedings of the 2008 IEEE International Conference on Robotics and Automation (ICRA)*, pages 1928–1935, 2008.
- [15] Jur van den Berg, Stephen J. Guy, Ming Lin, and Dinesh Manocha. Reciprocal n -body collision avoidance. In *Robotics Research: The 14th International Symposium ISRR*, volume 70 of *Springer Tracts in Advanced Robotics*, pages 3–19. Springer, 2011.
- [16] Kai Zhang, Ruizhen Gao, and Jingjun Zhang. Research on trajectory tracking and obstacle avoidance of nonholonomic mobile robots in a dynamic environment. *Robotics*, 9(3):74, 2020.
- [17] Run Mao, Hongli Gao, and Liang Guo. A novel collision-free navigation approach for multiple nonholonomic robots based on ORCA and linear MPC. *Mathematical Problems in Engineering*, 2020:4183427, 2020.
- [18] Javier Alonso-Mora, Andreas Breitenmoser, Martin Rufli, Paul Beardsley, and Roland Siegwart. Optimal reciprocal collision avoidance for multiple non-holonomic robots. In *Distributed Autonomous Robotic Systems*, volume 83 of *Springer Tracts in Advanced Robotics*, pages 203–216. Springer, 2013.
- [19] Javier Alonso-Mora, Paul A. Beardsley, and Roland Siegwart. Cooperative collision avoidance for nonholonomic robots. *IEEE Transactions on Robotics*, 34(2):404–420, 2018.
- [20] Zenan Lin, Ming Yue, Guangyi Chen, and Jianzhong Sun. Path planning of mobile robot with PSO-based APF and fuzzy-based DWA subject to moving obstacles. *Transactions of the Institute of Measurement and Control*, 44(1):121–132, 2022.
- [21] Diego Martinez-Baselga, Eduardo Sebastián, Eduardo Montijano, Luis Riazuelo, Carlos Sagüés, and Luis Montano. AVOCADO: Adaptive optimal collision avoidance driven by opinion. *IEEE Transactions on Robotics*, 41:2495–2511, 2025.
- [22] Diego Martinez-Baselga, Diego Lanaspá, Luis Riazuelo, and Luis Montano. Adaptive optimal collision avoidance of dynamic agents for differential-drive robots. *Robotics*, 15(4):72, 2026.

- [23] Nicolas Montés, Francisco Chinesta, Marta C. Mora, Antonio Falcó, Lucia Hilario, Nuria Rosillo, and Enrique Nadal. Real-time path planning based on harmonic functions under a proper generalized decomposition-based framework. *Sensors*, 21(12):3943, 2021.
- [24] Mark Moll, Constantinos Chamzas, Zachary Kingston, and Lydia E. Kavraki. HyperPlan: A framework for motion planning algorithm selection and parameter optimization. In *Proceedings of the 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2511–2518, 2021.
- [25] Shuai Liu and Pengcheng Liu. Benchmarking and optimization of robot motion planning with motion planning pipeline. *The International Journal of Advanced Manufacturing Technology*, 118:949–961, 2022.
- [26] Jing Xu, Wanruo Zhang, Jialun Cai, and Hong Liu. SafeCrowdNav: Safety evaluation of robot crowd navigation in complex scenes. *Frontiers in Neurobotics*, 17:1276519, 2023.
- [27] Stefan Mitsch, Khalil Ghorbal, David Vogelbacher, and André Platzer. Formal verification of obstacle avoidance and navigation of ground robots. *The International Journal of Robotics Research*, 36(12):1312–1340, 2017.
- [28] Gary Bradski. The OpenCV library. *Dr. Dobbs Journal of Software Tools*, 25(11):120–125, 2000.
- [29] Sergio Garrido-Jurado, Rafael Muñoz Salinas, Francisco José Madrid-Cuevas, and Manuel Jesús Marín-Jiménez. Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, 47(6):2280–2292, 2014.
- [30] Zhengyou Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000.
- [31] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, Cambridge, 2nd edition, 2004.
- [32] Israel M. Gelfand and Sergei V. Fomin. *Calculus of Variations*. Prentice-Hall, Englewood Cliffs, NJ, 1963.
- [33] Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyu Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995.
- [34] Pauli Virtanen et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [35] Jorge J. Moré, Burton S. Garbow, and Kenneth E. Hillstom. User guide for MINPACK-1. Technical Report ANL-80-74, Argonne National Laboratory, Argonne, IL, 1980.
- [36] Loup Verlet. Computer “experiments” on classical fluids. I. thermodynamical properties of Lennard-Jones molecules. *Physical Review*, 159(1):98–103, 1967.
- [37] John E. Jones. On the determination of molecular fields. II. from the equation of state of a gas. *Proceedings of the Royal Society of London A*, 106(738):463–477, 1924.
- [38] Michael D. McKay, Richard J. Beckman, and William J. Conover. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 21(2):239–245, 1979.

A Cost terms, gradient and auxiliary expressions

This appendix gives the expressions that the body of the paper states only by name. Terms already written in full in Sections 3.2 and 3.3 — the homography, the parallax factor, the smoothing penalty and the two kinematic penalties — are not repeated here.

A.1 Arena boundary: signed distances

The wall potential of Equation (12) is expressed on a signed distance to an effective boundary. For a point $\mathbf{x}$ and effective limits obtained by shrinking the physical bounds by the safety radius and the arena margin, the four signed distances are

$$s_L = x - x_{\min}^{\text{eff}}, \quad s_R = x_{\max}^{\text{eff}} - x, \quad s_B = y - y_{\min}^{\text{eff}}, \quad s_T = y_{\max}^{\text{eff}} - y, \quad (\text{A.1})$$

each positive inside the effective arena, zero on its limit and negative outside it. The four contributions are summed and weighted by λ_{ar} in the potential V of Equation (13). Because Equation (12) remains defined for $s < 0$, a trajectory that leaves the effective arena is penalised increasingly rather than rejected, which keeps the problem unconstrained.

A.2 Finite core penalty

The compact obstacle potential of Equation (11) is defined on the edge-to-edge clearance and vanishes at the influence radius. It must also remain finite when a candidate trajectory passes *inside* an obstacle, which a multi-start procedure will attempt; it is differentiable there except at the exact obstacle centre. Writing $\delta = \max(0, r_k + r_{\text{rob}} - \|\mathbf{x} - \mathbf{c}_k\|)$ for the penetration depth, the core contributes

$$V_{\text{core}}(\delta) = 1 + G \left(\frac{\delta}{\delta_0} \right)^2, \quad G = 100, \quad \delta_0 = 0.02 \text{ m}, \quad (\text{A.2})$$

in units of $\gamma g_{o,k}$. The additive unit makes the expression continuous with the compact potential, which reaches unity at contact. *No hard geometric constraint is imposed*: the optimisation remains unconstrained, and a trajectory crossing an obstacle is expensive rather than infeasible. This is what allows a straight-line initial guess to be used even when it is invalid.

A.3 Analytic gradient of the discrete action

The optimiser is supplied with the analytic gradient of Equation (13) with respect to the interior nodes $\mathbf{x}_1, \dots, \mathbf{x}_{N-1}$, the endpoints being fixed. It is exact wherever the cost is differentiable; the radial terms below are not differentiable at an obstacle centre, where the implementation returns a zero contribution by convention. Because the optimisation is unconstrained and obstacle clearance is not a multi-start rejection criterion, this convention is a numerical fallback, not a geometric guarantee. The gradient is the sum of four contributions, assembled node by node.

The kinetic term couples each node to its two neighbours through the second difference,

$$\frac{\partial}{\partial \mathbf{x}_k} \sum_j \frac{\alpha \Delta t}{2} \left\| \frac{\mathbf{x}_{j+1} - \mathbf{x}_j}{\Delta t} \right\|^2 = -\frac{\alpha}{\Delta t} (\mathbf{x}_{k+1} - 2\mathbf{x}_k + \mathbf{x}_{k-1}). \quad (\text{A.3})$$

The potential is evaluated at segment midpoints, so each midpoint gradient is distributed to the two nodes that define it with weight one half:

$$\begin{aligned} \frac{\partial}{\partial \mathbf{x}_k} \sum_j \Delta t V \left(\frac{\mathbf{x}_{j+1} + \mathbf{x}_j}{2} \right) &= \frac{\Delta t}{2} [\nabla V(\mathbf{m}_{k-1}) + \nabla V(\mathbf{m}_k)], \\ \mathbf{m}_j &= \frac{\mathbf{x}_{j+1} + \mathbf{x}_j}{2}. \end{aligned} \quad (\text{A.4})$$

Within ∇V , each obstacle contributes radially. Outside the core, the compact polynomial of Equation (11) gives

$$\begin{aligned} \nabla V_k(\mathbf{x}) &= -\frac{2p\gamma g_{o,k}}{R^2} \left(1 - \frac{d_k^2}{R^2} \right)^{p-1} d_k \hat{\mathbf{u}}_k, \\ \hat{\mathbf{u}}_k &= \frac{\mathbf{x} - \mathbf{c}_k}{\|\mathbf{x} - \mathbf{c}_k\|}. \end{aligned} \quad (\text{A.5})$$

and inside it the derivative of Equation (A.2), which is linear in the penetration and therefore bounded. The wall terms differentiate Equation (12) along the corresponding coordinate axis, with the sign of the outward normal.

The smoothing penalty differentiates to a five-point stencil, since each node appears in three consecutive second differences, and the two kinematic penalties differentiate through the chain rule applied to the squared discrete velocities and accelerations, both of which are themselves linear in the nodes.

The motivation is quantitative. With $2(N - 1)$ design variables, a gradient computed by forward differences costs one additional full cost evaluation per variable after the reference evaluation, and each cost evaluation sweeps every obstacle at every midpoint. The analytic gradient was validated against centred finite differences in a dedicated test.

A.4 Calibration score terms

The seven weighted terms of Equation (30) are built from fixed thresholds rather than from the population of profiles. Three of them measure a minimum clearance m against a target $q = 0.08$ m through

$$\Pi(m; q) = \begin{cases} 4 & \text{if } m \leq 0, \\ \frac{q - m}{q} & \text{if } 0 < m < q, \\ 0 & \text{if } m \geq q \text{ or } m \text{ undefined,} \end{cases} \quad (\text{A.6})$$

so that a clearance at or above the target costs nothing, a smaller clearance costs proportionally more, and a negative clearance saturates. Three further terms are linear ratios saturated at four — the mean final distance to the goal against 0.20 m, the mean acceleration saturation rate against 0.12, and the mean tracking error against 0.50 m.

The last term penalises the commanded force. For a single run r , ϱ_r denotes the ninety-fifth percentile over the run of the ratio $\|\mathbf{F}\| / (m a_{\max})$. Let $\mathcal{R}_F$ be the set of scored runs for which this ratio is finite. The implementation applies the floor $\varepsilon_F = 10^{-12}$ before taking the logarithm and computes

$$\ell_r = \log_{10}(\max(\varrho_r, \varepsilon_F)), \quad \bar{\ell} = \frac{1}{|\mathcal{R}_F|} \sum_{r \in \mathcal{R}_F} \ell_r.$$

Thus the logarithm is taken before the mean, not after. The penalty is measured against a reference ratio of 1.5:

$$\Pi_F = \begin{cases} 5 & \text{if } \mathcal{R}_F = \emptyset, \\ \min(5, \max(0, \bar{\ell} - \log_{10} 1.5)) & \text{otherwise.} \end{cases} \quad (\text{A.7})$$

The treatment of missing values is deliberately asymmetric and is stated here because it affects the ranking. Non-finite observations are omitted from each profile-level minimum or mean. If no finite value remains for a distance, saturation-rate or tracking-error aggregate, that aggregate contributes zero; if no finite force ratio remains, the force term takes its maximum penalty. The per-run diagnostic score uses the same zero versus maximum defaults when the corresponding observation is absent (represented by a missing value).

A.5 Command smoothing

The two wheel commands of Equation (28) are smoothed before transmission by a first-order filter whose coefficient is rescaled

to the elapsed control period, so that a variable loop rate does not change the effective time constant. With Δt_c the elapsed period and $\Delta t_{\text{ref}} = 0.08$ s the reference period,

$$\begin{aligned} \alpha_{\text{eff}} &= 1 - (1 - \alpha_{\text{cmd}})^{\Delta t_c / \Delta t_{\text{ref}}}, \\ u^{(n)} &= \alpha_{\text{eff}} u_{\text{cmd}}^{(n)} \\ &\quad + (1 - \alpha_{\text{eff}}) u^{(n-1)}. \end{aligned} \quad (\text{A.8})$$

applied independently to each wheel. The filter history is cleared whenever the direction of travel changes, so that a command of the previous direction cannot survive a transition; the mechanism is described in Section 3.5.2.

A.6 Exclusive decomposition of run outcomes

The four categories used in Section 3.7 are defined so as to be mutually exclusive and exhaustive. With I_s and I_g the indicators of Equations (2) and (3), and E the indicator of an execution error,

$$\begin{aligned} C_1 &= I_s I_g, & C_2 &= I_s (1 - I_g), \\ C_3 &= (1 - I_s)(1 - E), & C_4 &= (1 - I_s)E. \end{aligned} \quad (\text{A.9})$$

with $C_1 + C_2 + C_3 + C_4 = 1$ for every run, since $I_s = 0$ whenever $E = 1$ by the definition of Equation (2). Figure 11 displays only the first three categories because no execution error was observed in the plotted data, so $C_4 = 0$. Category C_2 — safe but short of the goal — includes the progression failure cases discussed in Section 4.1.2; the indicator alone does not assign that cause to every terminal miss.